

PRIMARY 4 BASIC DIGITAL LITERACY SCHEME OF WORK

FIRST TERM

Week	Topic	Content
1	Introduction to Digital Devices	Understanding technology and digital devices. Types: computers, tablets, smartphones, smartwatches, cameras. Computer types: desktop, laptop, tablet. Basic parts: monitor, keyboard, mouse, CPU. Uses in daily life. Care and handling of devices. Computer lab safety rules.
2	History of Digital Devices	Evolution from abacus to modern computers. Five generations: vacuum tubes, transistors, integrated circuits, microprocessors, AI. Key inventors: Charles Babbage, Ada Lovelace. Changes in size, speed, and cost over time.
3	Parts of Digital Devices I	Hardware vs Software basics. Input devices: keyboard, mouse, microphone, scanner, webcam. Output devices: monitor, printer, speakers, headphones. Functions of each device. Proper use and care.
4	Parts of Digital Devices II	Internal parts: CPU (brain), RAM (temporary memory), Hard Drive/Storage (permanent memory), Motherboard (connects everything). Power supply and cooling systems. How parts work together.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Peripheral Devices	Definition: external devices connected to computer. Common peripherals: printer, scanner, external hard drive, USB flash drive, webcam, speakers. Connecting devices using USB, HDMI, Bluetooth. Uses of each

		peripheral device.
8	Input Devices	Detailed study of input devices. Keyboard: QWERTY layout, typing practice, special keys. Mouse: left/right click, scroll, drag and drop. Touchpad and touchscreen. Microphone for voice input. Scanner for documents and images.
9	Output Devices	Detailed study of output devices. Monitor: displaying information, screen resolution. Printer: types (inkjet, laser), printing documents. Speakers and headphones: audio output. Projector for presentations.
10	Computer Care and Safety	Proper handling: don't drop, keep away from water and food. Cleaning: dust removal, screen cleaning, keyboard cleaning. Virus protection basics. Power management: proper shutdown, using surge protectors. Ergonomics: proper sitting position, screen height, keyboard position. Lab safety rules review.
11	REVISION	Review of all topics covered in First Term. Practical identification of computer parts.
12	EXAMINATION	End of First Term Examination (Theory and Practical)
13	CLOSING	Report distribution and vacation

PRIMARY 4 BASIC DIGITAL LITERACY LESSON NOTES

FIRST TERM

WEEK 1: INTRODUCTION TO DIGITAL DEVICES.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define the term "Digital Device."
- List three examples of digital devices used at home or school.
- Identify the basic parts of a computer system.
- State three safety rules to follow in the computer lab.

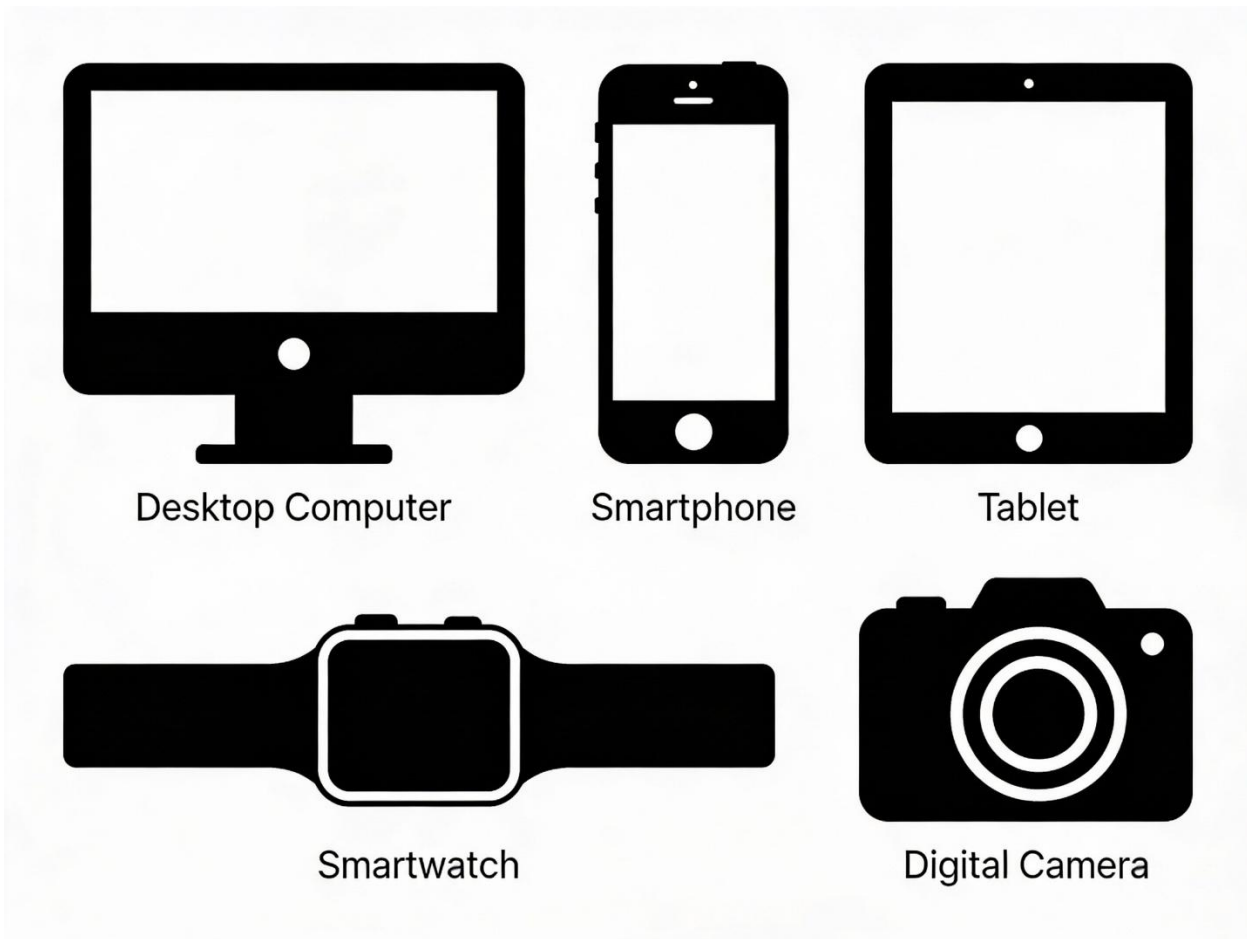
CONTENT

Understanding Digital Devices: A digital device is an electronic machine that works with information. It can receive information, store it, change it, and send it to another place. Think of it like a smart helper. Just like you need food for energy, digital devices need electricity from a wall plug or power from batteries to wake up and work. They make jobs like writing, calculating, and talking to people far away much quicker and easier.

Types of Digital Devices: We See Every Day: We are surrounded by digital devices. They come in different shapes and sizes for different jobs.

- **Computers:** These are like super-smart boxes. We use them for typing our classwork, learning new things from educational programs, and exploring the internet to see pictures and videos from all over the world.
- **Smartphones:** These are telephones that can do many other things. You can use them to take photographs of your family, play fun games, watch cartoons, and call your mummy or daddy.
- **Tablets:** These look like big, flat smartphones. They are good for reading digital storybooks, drawing pictures with your finger, and watching video lessons. They are easy to carry.
- **Smartwatches:** These are worn on the wrist like a normal watch, but they can also count how many steps you walk, show messages from your parents, and tell you the time.

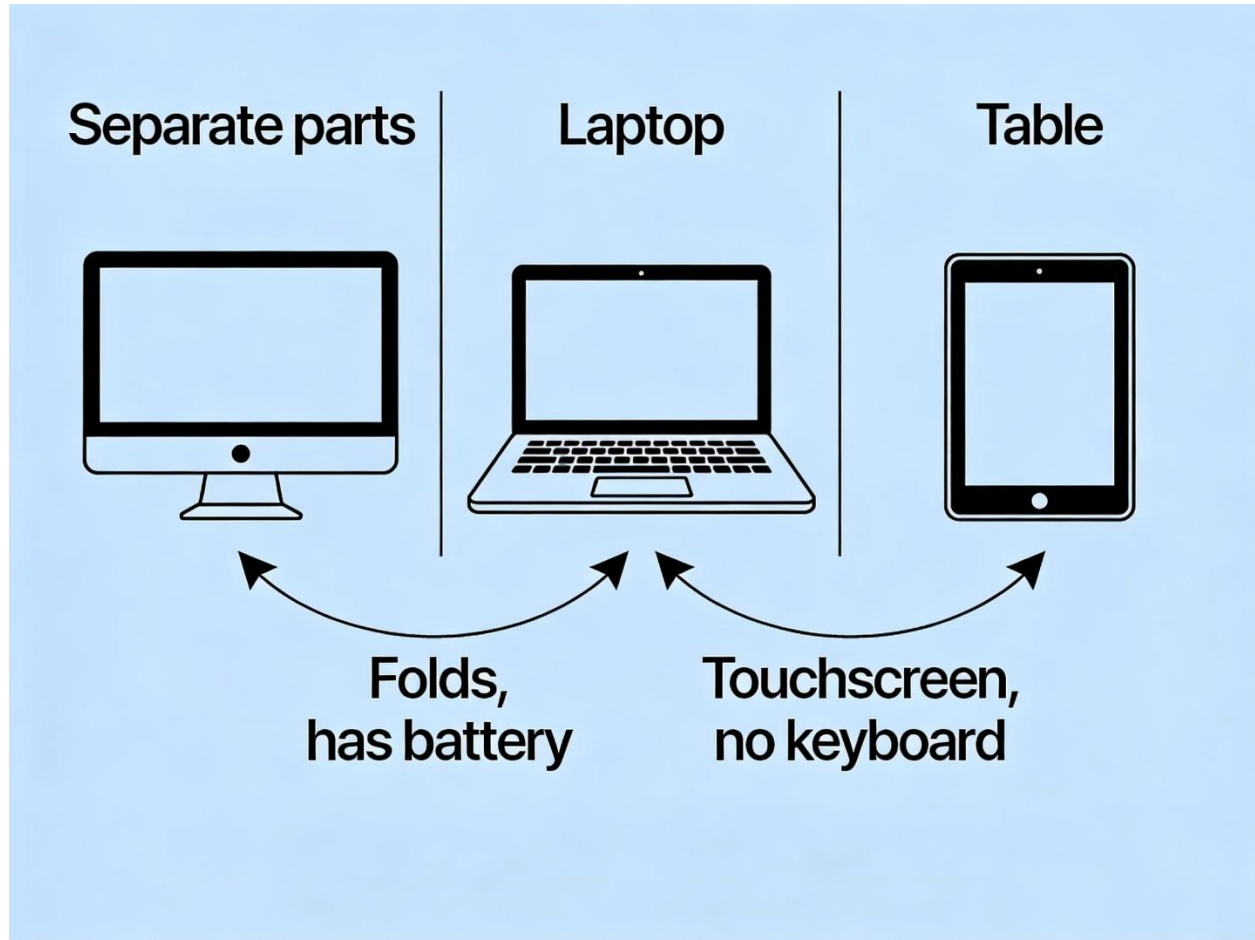
- **Digital Cameras:** These are used to take very clear pictures and record videos, like at a birthday party or school graduation.



Types of Computers: Not all computers look the same. Here are the main ones:

- **Desktop Computer:** This is a big computer that usually stays in one place, like on a desk in the computer lab or in an office. It has many separate parts: a screen (monitor), a typing board (keyboard), a pointing device (mouse), and a big box (system unit) that are all connected by wires.
- **Laptop:** This is a portable computer. You can fold it like a book to carry it. It has a battery inside, so it can work without being plugged into electricity for some time. It has the screen, keyboard, and system unit all built into one.

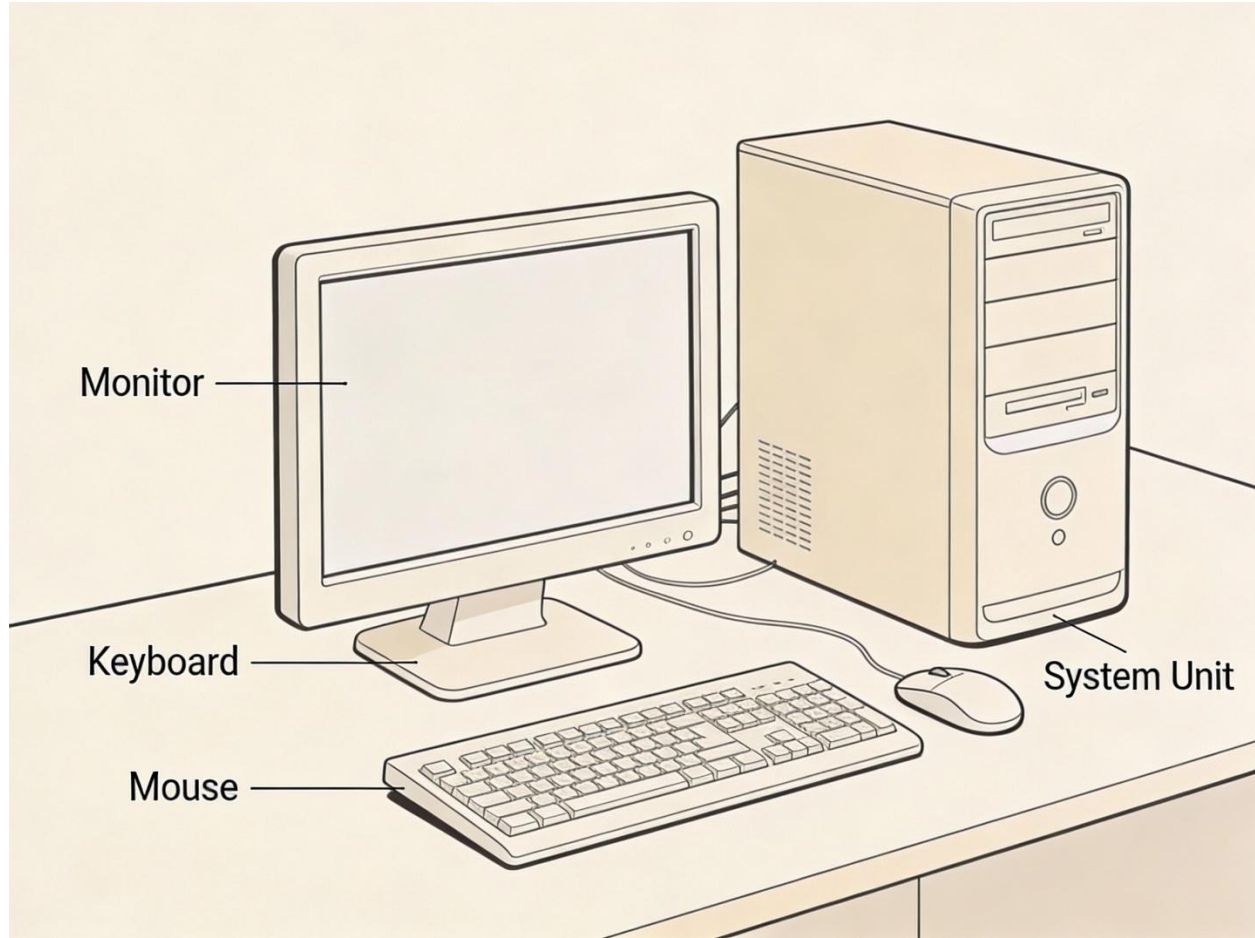
- **Tablet:** This is a flat, slate-like computer. It has a touchscreen, which means you use your fingers to tap and swipe on the screen instead of using a mouse. It does not have a physical keyboard you can click; you use a keyboard that appears on the screen.



Basic Parts of a Computer (Using a Desktop as our example): Let's look at the main parts you can see and touch:

- **Monitor:** This looks like a television. It is the screen that shows you everything you are doing on the computer—your words, your drawings, and videos.
- **Keyboard:** This has many buttons called "keys." You press these keys to type letters, numbers, and symbols into the computer.
- **Mouse:** This is a small device you move with your hand on the table. It has buttons on top. It controls a little arrow or pointer on the monitor. You use it to select things, open programs, and click on buttons.

- **System Unit (CPU Case):** This is the big box that holds the brain and heart of the computer. All the wires from the monitor, keyboard, and mouse connect to it. It powers the whole system.



Computer Lab Safety Rules: The computer lab has expensive and delicate equipment. We must follow rules to protect ourselves and the computers.

- **No Food or Drink:** Never eat biscuits or drink water near the computer. Crumbs can fall into the keyboard and spoil it, and water can spill and cause a dangerous electric shock or break the computer.
- **No Running:** Always walk slowly and carefully in the lab. If you run, you might trip over wires or knock a computer off the table.
- **Hands Off Wires:** Never touch loose or broken wires or cables. They can give you a serious electric shock. Always tell your teacher if you see a damaged wire.

- **Clean Hands:** Always wash your hands with soap and dry them completely before using the computer. Dirty, sticky hands can make the keyboard and mouse greasy and dirty.

EVALUATION

- What is a digital device? (An electronic machine that needs electricity to process, store, and send information).
- Give two differences between a desktop computer and a laptop. (A desktop has separate parts and stays in one place; a laptop is all-in-one, folds, and can be carried around).
- Which part of the computer displays pictures and words like a television? (The Monitor).
- Why is it dangerous to drink water near a computer? (It can spill, cause an electric shock, or damage the computer parts).
- Look around your home. Name three digital devices you can see. (e.g., Television, Smartphone, Calculator).

ASSIGNMENT

- (Drawing Practical) Draw and colour a simple picture of a computer monitor.
- (Oral/Written) Tell your parent or guardian two ways your family uses digital devices at home (e.g., Daddy uses his phone for work calls, Mummy uses a laptop to check prices in the market). Write them down.
- (Written) Write down one important safety rule for the computer lab and explain why it is important.

WEEK 2: HISTORY OF DIGITAL DEVICES.

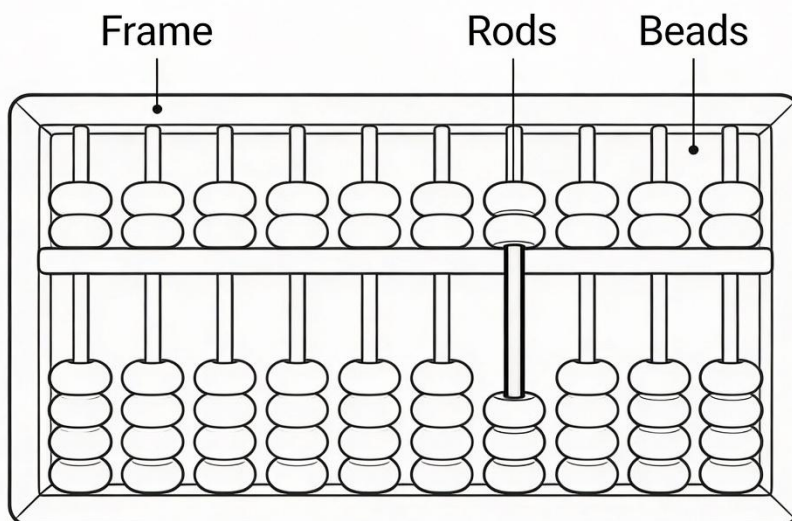
OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify the Abacus as an early counting device.
- Mention the "Father of Computers."
- List the five generations of computers.
- Describe how computers have changed in size and speed over time.

CONTENT

Early Counting Devices: Long, long ago, before there were any computers or even calculators, people still needed to count things—like their animals, money, or goods in the market. They used their fingers first, but then created tools to help with bigger numbers.

- **The Abacus:** This was one of the very first counting machines, invented thousands of years ago. It is a wooden frame with beads that slide on rods. By moving the beads up and down, people could do maths like addition and subtraction very quickly. It is still used in some places today to teach children about numbers.



Key Inventors in Computer History: Two very important people helped start the idea of the modern computer.

- **Charles Babbage:** He is called the "Father of the Computer." He was an English inventor who, about 200 years ago, designed a machine called the "Analytical Engine." This machine could follow a set of instructions to solve maths problems. Although it was never fully built during his lifetime, his ideas were the blueprint for today's computers.
- **Ada Lovelace:** She was a mathematician who worked with Charles Babbage. She wrote the first-ever set of instructions (a program) for the Analytical Engine to follow. Because of this, she is known as the world's *first computer programmer*.



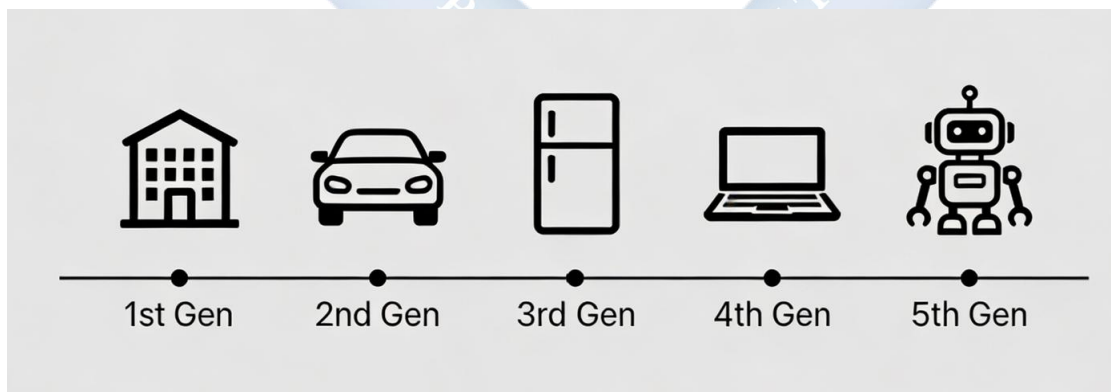
Charles Babbage
Father of the Computer



Ada Lovelace
First Computer Programmer

The Five Generations of Computers: Computers didn't become small and fast overnight. They improved in big steps, called "Generations." Each generation used a new, better technology.

- **First Generation (Vacuum Tubes):** These computers, made around the 1940s, were as big as a whole classroom! They used parts called vacuum tubes, which got very hot and used a lot of electricity. They were also very slow and could only do simple calculations.
- **Second Generation (Transistors):** In the 1950s, scientists invented the transistor. These were smaller, cooler, and more reliable than vacuum tubes. Computers became smaller (about the size of a car) and much faster.
- **Third Generation (Integrated Circuits):** In the 1960s, many tiny transistors were put together on a small chip called an "Integrated Circuit" or "microchip." This made computers smaller (like a fridge), cheaper, and even faster.
- **Fourth Generation (Microprocessors):** From the 1970s till now, the microprocessor was invented. It put the entire brain of the computer (the CPU) on a single, tiny chip. This is the technology in the laptops, desktops, and tablets we use today. They are small, powerful, and affordable.
- **Fifth Generation (Artificial Intelligence - AI):** This is the present and future. These computers can learn and make decisions almost like humans. Examples are robots that can clean houses, voice assistants that answer your questions, and apps that suggest videos you might like.



Evolution of Size and Speed

- **Then (Past):** The first computers were *Gigantic, Slow, and Costly*. They filled entire rooms, were slower than today's calculators, and only very rich companies or governments could afford them.

- **Now (Present):** Today's computers are *Small, Fast, and Common*. You can hold a powerful smartphone in your palm. They can do billions of calculations in one second. Many families can afford at least one computer or smartphone.

EVALUATION

- What was the Abacus used for? (For counting and doing calculations like addition and subtraction).
- Why is Charles Babbage famous? (He designed the first mechanical computer, so he is called the Father of the Computer).
- We use laptops and smartphones today. Which generation of computers do they belong to? (Fourth Generation - Microprocessors).
- Who was Ada Lovelace, and what did she do? (She was the first computer programmer; she wrote instructions for Babbage's machine).
- True or False: The first computers were so small you could carry them in your school bag. (False. They were as big as a room).

ASSIGNMENT

- (Written) Write the full name of the "Father of Computers" (Charles Babbage).
- (Drawing Practical) Draw a simple picture of an Abacus.
- (Oral/Written) Tell a story to your friend or write one sentence: "Computers have changed from being as big as a _____ to being as small as a _____." (e.g., room, palm).

WEEK 3: PARTS OF DIGITAL DEVICES I.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Differentiate between Hardware and Software.
- Define Input Devices and give examples.
- Define Output Devices and give examples.
- Match devices to their functions.

CONTENT

Hardware vs. Software: The Body and the Mind: Think of a computer like a person.

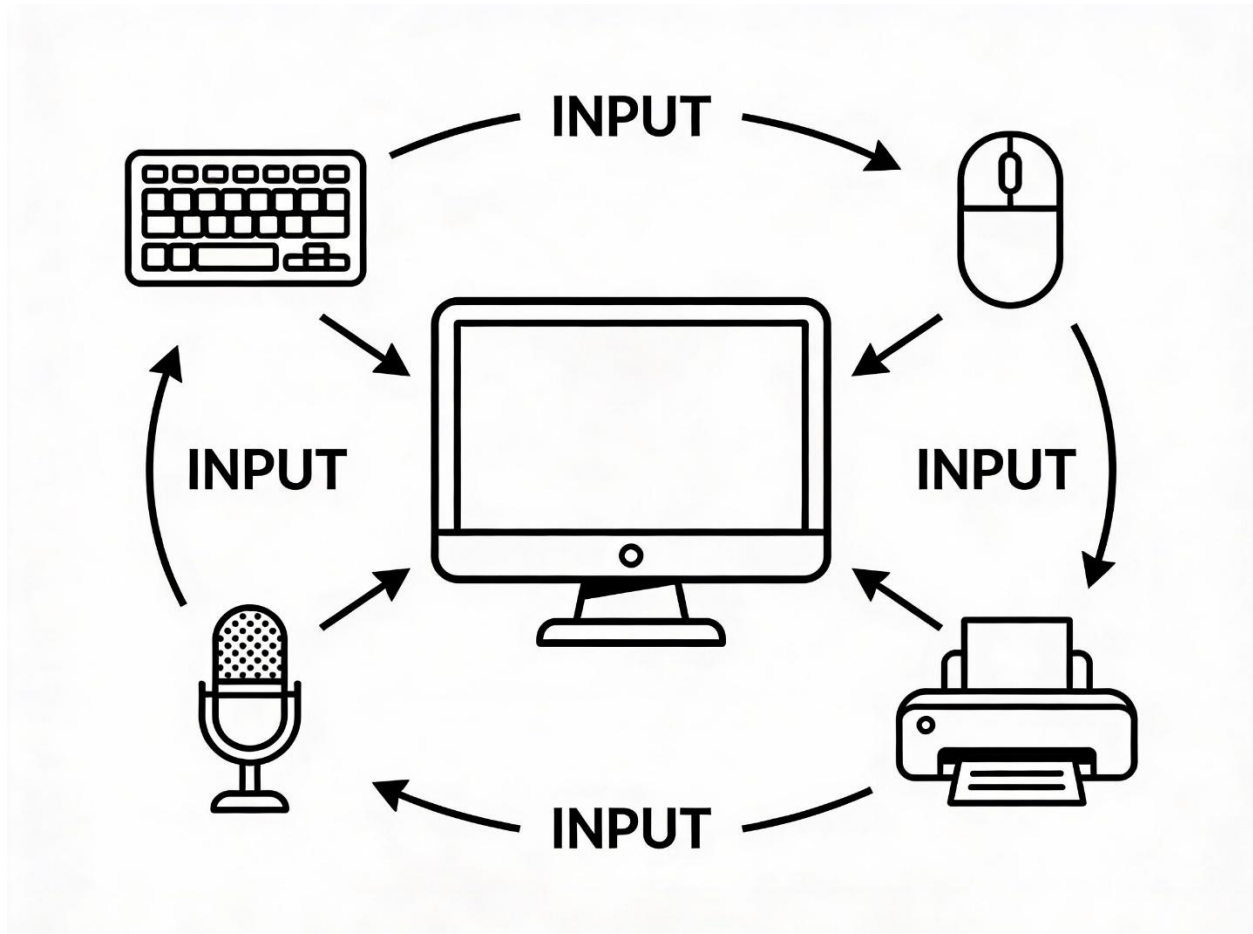
- **Hardware:** This is the *body* of the computer—all the physical parts you can see, touch, and feel. *If you can knock on it, it's hardware.* Examples: The plastic case, the glass screen, the keyboard keys, the mouse, the speakers, and the wires.
- **Software:** This is the *mind* or the *knowledge* inside the computer. You cannot touch it. It is the set of instructions and programs that tell the hardware what to do. Examples: The game you play, the program you use to type (like Microsoft Word), the app you use to draw, and even the computer's main operating system (like Windows).

Hardware	Software
You can TOUCH it.	You CANNOT touch it.
It is PHYSICAL.	It is a PROGRAM.
Example: Computer mouse	Example: Game on the computer

Input Devices: Bringing Information IN: Input devices are like our eyes, ears, and mouth. They are used to send (input) information *INTO* the computer. They help us "talk to" the computer.

- **Keyboard:** You type letters and numbers. It sends the words you type into the computer.
- **Mouse:** You move it and click its buttons. It sends commands like "open this" or "select that" into the computer.

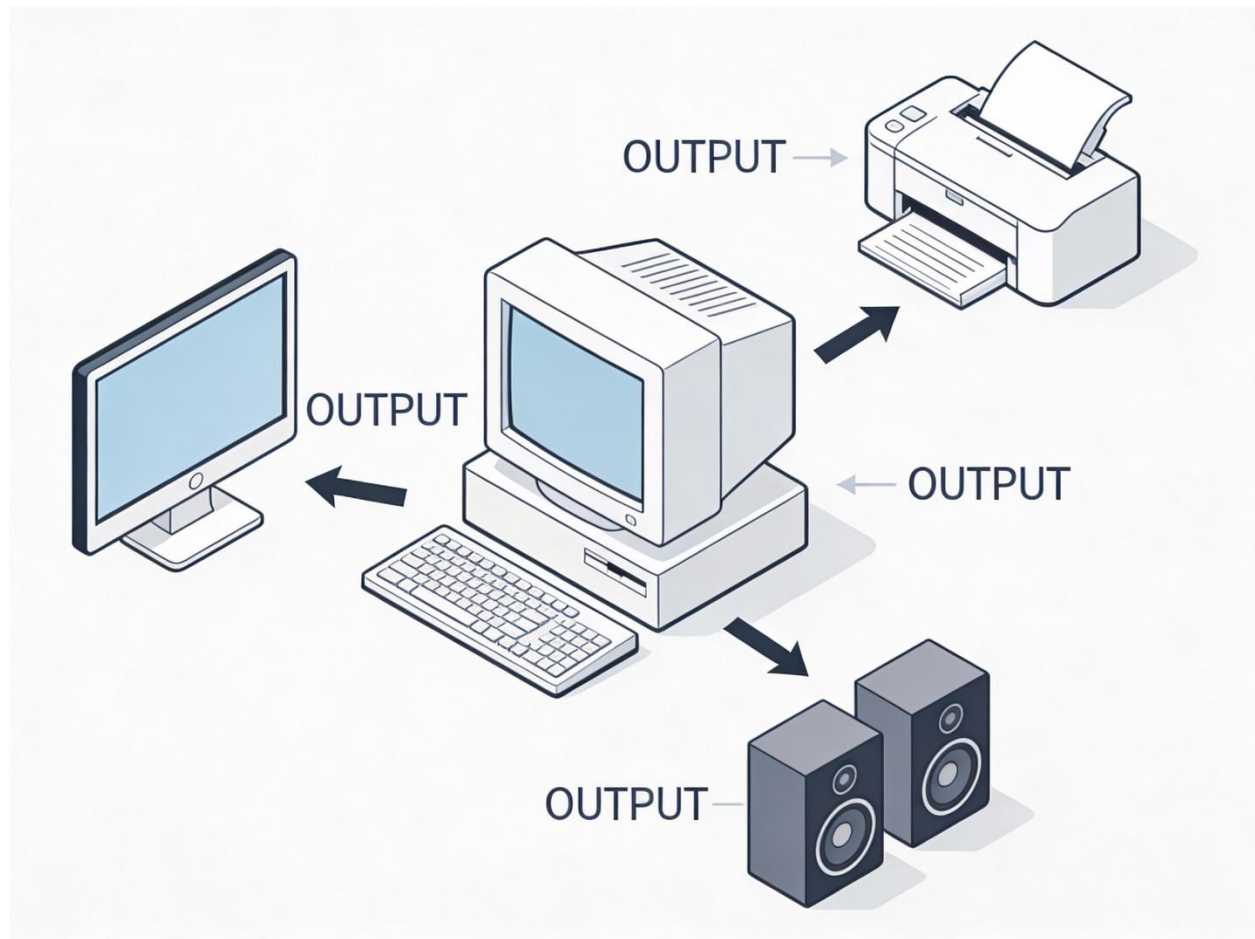
- **Microphone:** You speak into it. It sends your voice or any sound into the computer so it can record or understand you.
- **Webcam:** It is a small camera on top of the monitor. It takes pictures or videos of you and sends them into the computer for video calls.
- **Scanner:** It works like a photocopier. You put a paper photo or document inside, and it sends a digital copy of that paper into the computer.



Output Devices: Getting Information OUT: Output devices are like our mouth and actions. They are used to get (output) information *OUT* of the computer. They help the computer "talk to" us or show us results.

- **Monitor/Screen:** It shows us the work, pictures, and videos from the computer. This is how we see what's happening.

- **Printer:** It takes the work from the computer and prints it on paper. This gives us a "hard copy" we can hold and carry.
- **Speakers/Headphones:** They let us hear the sounds, music, or voices coming from the computer.



EVALUATION

- Can you touch and hold computer software in your hand? (No, software is a program, not a physical object).
- A mouse helps you select things on the screen. Is it an Input or Output device? (Input device).
- You want to listen to a story from the computer without disturbing others. What output device should you use? (Headphones or Speakers).

- What is one main difference between hardware and software? (Hardware is physical/touchable; software is not).
- You want to write your name on the computer screen. Which input device will you use? (Keyboard).

ASSIGNMENT

- (Written) Look around your house. List three things that are computer Input devices.
- (Written) List three things that are computer Output devices.
- (Drawing Practical) Draw a pair of headphones and label it as an "Output Device."



WEEK 4: PARTS OF DIGITAL DEVICES II.

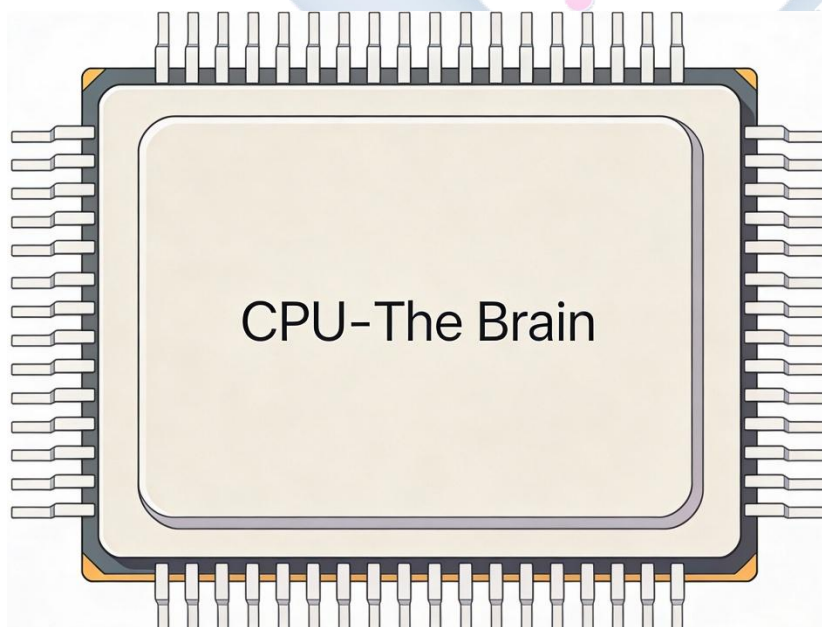
OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify the internal parts of a computer.
- Explain the function of the CPU.
- Differentiate between RAM (temporary memory) and Hard Drive (permanent memory).
- Describe the role of the Motherboard.

CONTENT

Internal Parts of a Computer: Inside the system unit (the big box of a desktop computer) are very important parts we don't usually see. We must never open the computer ourselves, but it's good to know what's inside. It's like knowing the engine is inside a car's bonnet.

The CPU (Central Processing Unit): The Brain The *CPU* is the *brain* of the computer. It is a small, flat chip (smaller than your palm). Its job is to think, calculate, and control everything. When you press a key or click the mouse, the CPU gets the message, decides what to do, and tells other parts what to do. It works incredibly fast, doing millions of calculations every second.



Memory: RAM vs. Hard Drive (Storage): A computer has two main types of memory, just like you have two types of memory:

- **RAM (Random Access Memory) - Your Temporary Memory or Workbench:** This is where the computer keeps the information for the work you are doing *right now*. For example, the story you are typing, the game you are playing. It is *very fast* so the CPU can get information quickly. It is *temporary*. When you turn off the computer, the RAM forgets everything. It's like a whiteboard you wipe clean.
- **Hard Drive (or Storage) - Your Permanent Memory or Cupboard:** This is where the computer saves your files *forever*, so you can find them later. Your saved pictures, videos, music, and documents are kept here. It is *permanent*. Even when the computer is turned off for days, your files are safe on the Hard Drive. It's like a cupboard or a box where you keep your toys.

	RAM (Workbench)	Hard Drive (Cupboard)
Purpose	For work you are doing NOW	For saving files FOREVER
Speed	Very Fast	Slower than RAM
When power is off	Forgets everything	Remembers everything

The Motherboard: The Backbone The *Motherboard* is a large, flat, green electronic board inside the system unit. It is like the *skeleton* of the computer or a *city's road network*. Its job is to connect all the important parts (CPU, RAM, Hard Drive) together and let them send messages to each other. Everything plugs into the Motherboard.

Power Supply Unit: This part takes electricity from the wall socket and changes it to the right type and amount needed by the CPU, motherboard, and other parts. It gives them power to work.

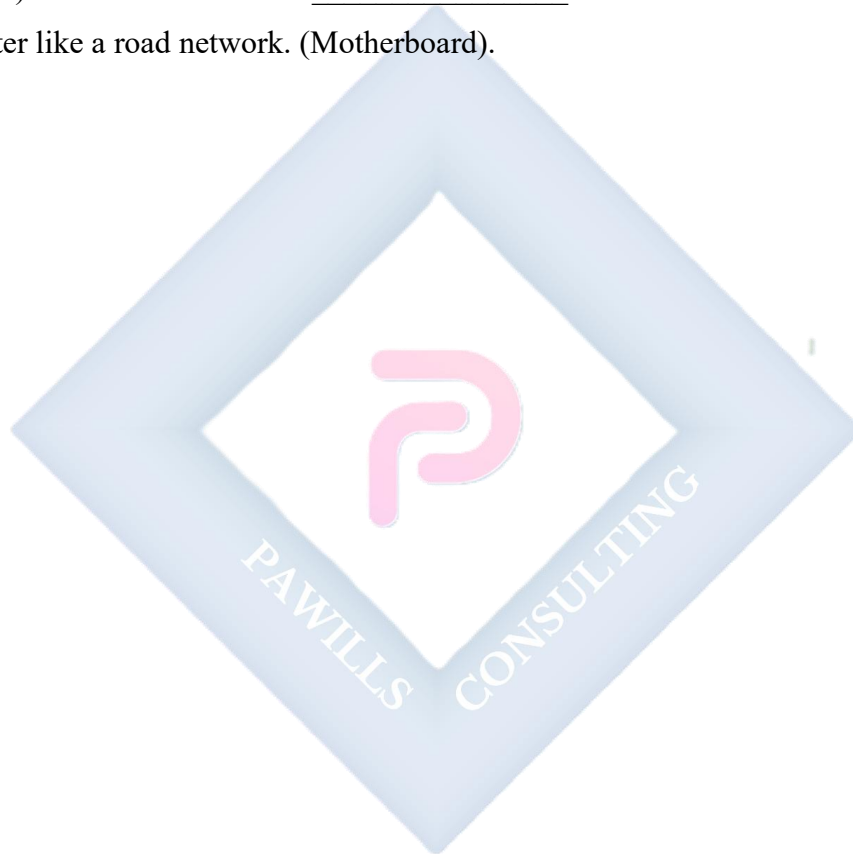
EVALUATION

- Which internal part acts as the computer's brain, doing all the thinking? (The CPU).
- You are typing a story but haven't saved it. The lights go out. Why does the computer forget your story? (Because it was only in the temporary RAM memory).
- Where does the computer keep your family photos so you can look at them next week? (On the Hard Drive / Storage).
- What is the job of the Motherboard? (To connect all the internal parts together so they can communicate).

- What does "CPU" stand for? (Central Processing Unit).

ASSIGNMENT

- (Drawing Practical) Draw a simple square and label it "CPU - The Brain of the Computer."
- (Written) In your own words, write one sentence to explain what the Hard Drive does. (e.g., "It is where the computer stores my files to keep them safe.").
- (Written) Fill in the blank: The _____ connects all the internal parts of the computer like a road network. (Motherboard).



WEEK 5: MIDTERM EXAMINATION.

(Assessment covering topics from Weeks 1-4)



WEEK 6: MIDTERM BREAK.

(Holiday/Rest Period)



WEEK 7: PERIPHERAL DEVICES.

OBJECTIVES By the end of the lesson, pupils should be able to:

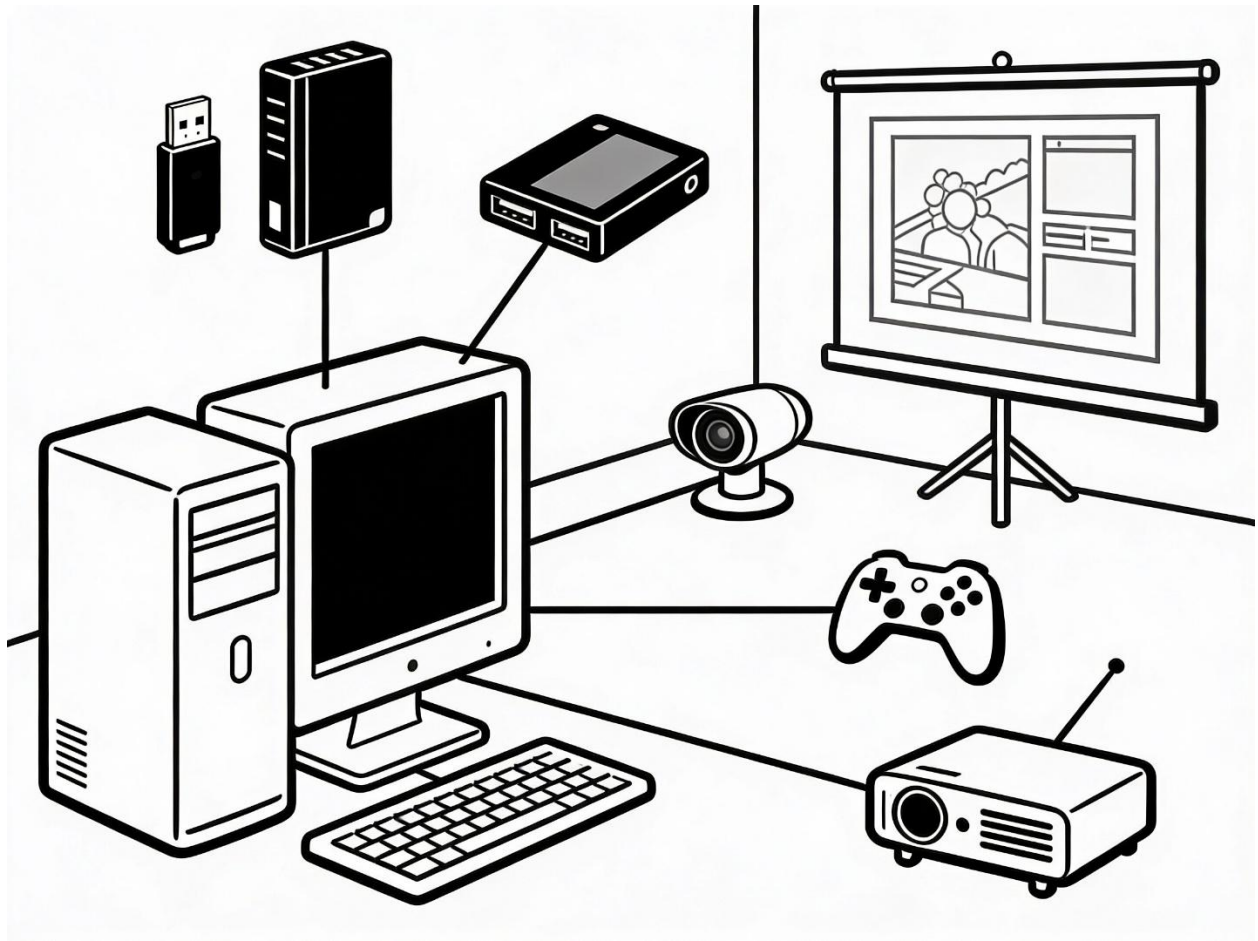
- Define Peripheral Devices.
- List four examples of peripheral devices.
- Identify ways to connect peripherals (USB, Bluetooth).
- State the use of a flash drive.

CONTENT

Definition of Peripheral Devices: Peripheral devices are *extra tools or accessories* that we connect to the main computer (like a desktop or laptop) from the *outside*. They are not the main computer itself, but they add new abilities to it, making it more useful. Think of them like adding a trailer to a truck or a new brush to a painter's set.

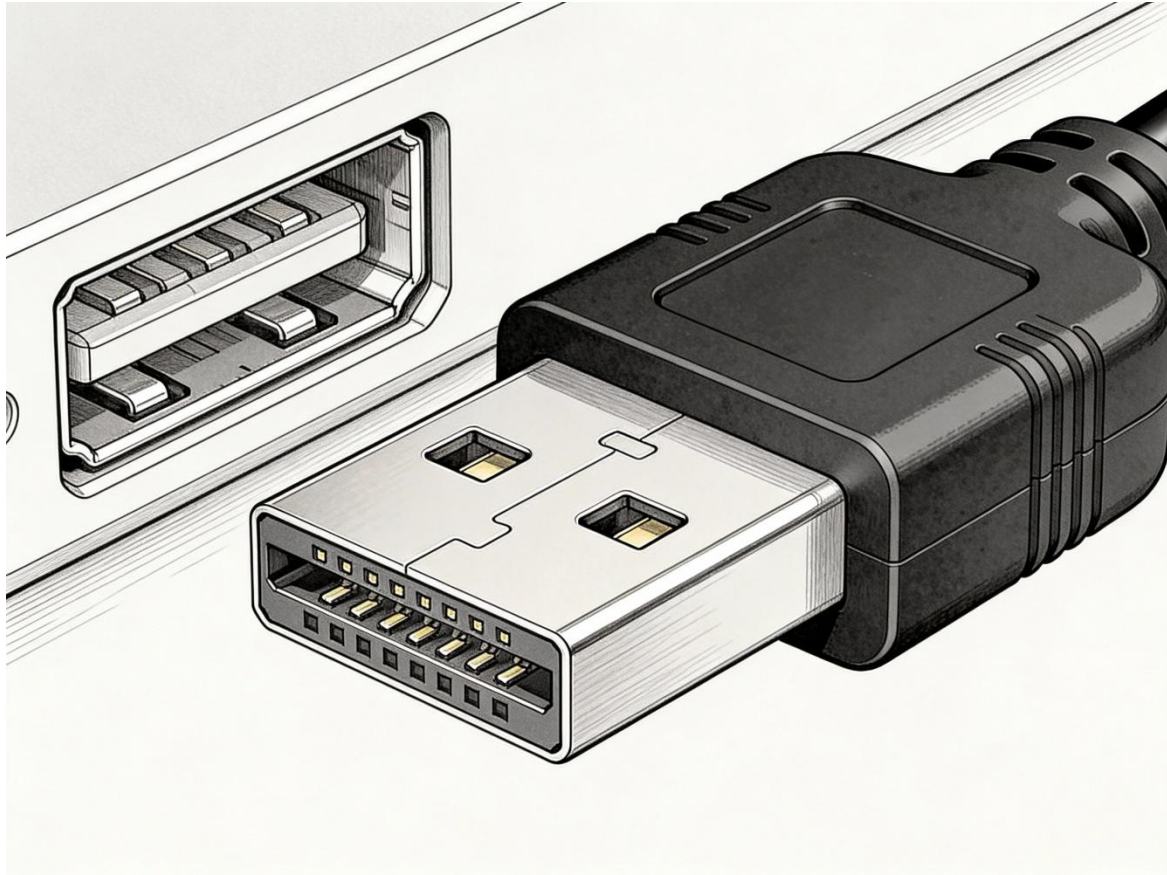
Common Peripheral Devices:

- **USB Flash Drive (Pen Drive):** A very small, portable stick. It is used to save files (like homework, pictures) and carry them from one computer to another. It plugs into a USB port.
- **External Hard Drive:** Looks like a small box. It does the same job as a flash drive but can store much, much more information—like thousands of movies or pictures.
- **Webcam:** A small camera that sits on top of your monitor. While some laptops have them built-in, you can add a separate one to a desktop computer for clearer video calls.
- **Joystick/Game Controller:** Makes playing computer games more fun and easier to control than using a keyboard.
- **Projector:** Takes the image from the computer screen and shows it very large on a wall or white screen. Teachers use it in class so everyone can see.



How We Connect These Devices: Peripherals need a way to "talk" to the computer. We connect them using:

- **USB Port and Cable:** This is the most common way. A USB port is a small, rectangular hole on the computer. We use a USB cable to plug in devices like the mouse, keyboard, flash drive, or printer.



- **HDMI Cable:** This is a wider, flat cable. It is used to connect a computer to a bigger screen like a Television or a Projector to show videos.
- **Bluetooth:** This is a *wireless* way to connect. It uses invisible radio waves. Devices like wireless mice, wireless headphones, or speakers can connect to the computer without any cables, as long as both have Bluetooth.

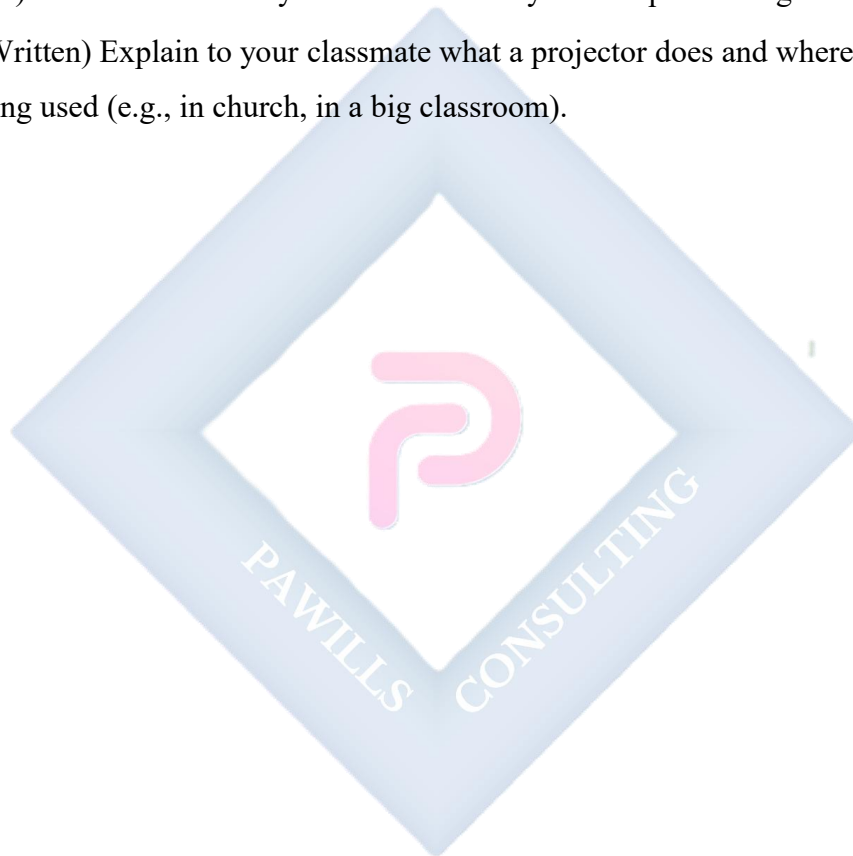
EVALUATION

- What do we call extra devices we connect to a computer to give it more functions? (Peripheral Devices).
- Name two peripheral devices used for storing and moving files. (USB Flash Drive, External Hard Drive).
- What is the main job of a USB Flash Drive? (To save files and carry them between computers).

- If you want to connect headphones to a computer without any wires, what technology would you use? (Bluetooth).
- You want to play a racing game on the computer. Which peripheral would make it more fun to control the car? (A Joystick or Game Controller).

ASSIGNMENT

- (Drawing Practical) Draw a picture of a USB Flash Drive.
- (Written) Name two devices you can connect to your computer using a USB cable.
- (Oral/Written) Explain to your classmate what a projector does and where you have seen one being used (e.g., in church, in a big classroom).



WEEK 8: INPUT DEVICES.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify the "QWERTY" keyboard layout.
- Demonstrate the actions of a computer mouse (click, drag).
- Explain the function of a scanner.
- Describe how a touchscreen works.

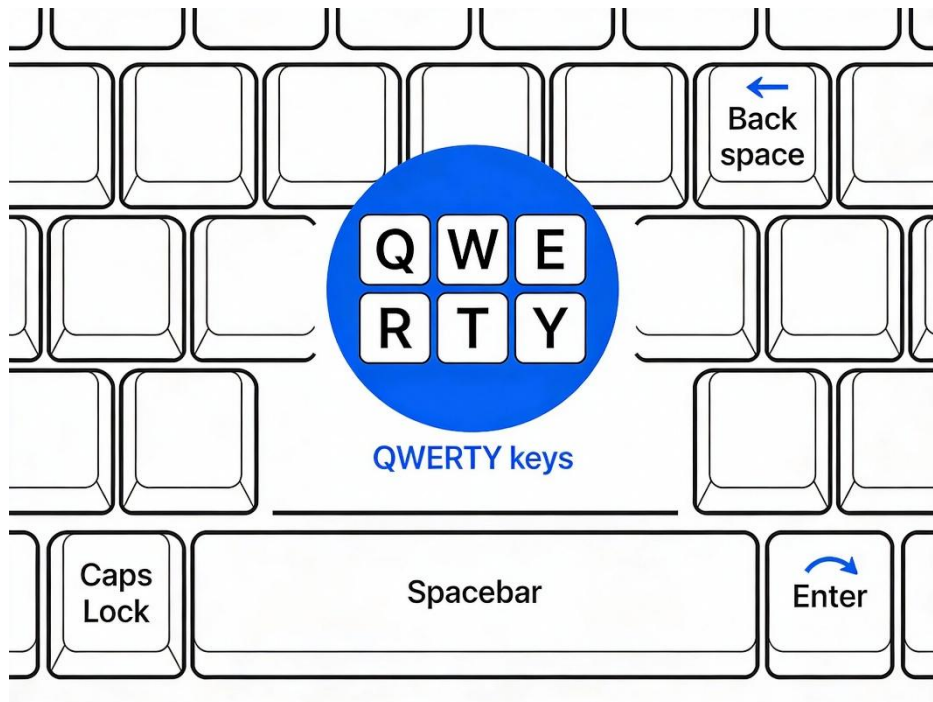
CONTENT

The Keyboard: Our Main Typing Tool The keyboard is the primary tool for putting words and numbers into the computer.

- **QWERTY Layout:** Look at the top row of letter keys. The first six letters are Q-W-E-R-T-Y. That's why it's called a "QWERTY" keyboard. This layout is used all over the world.

Special Keys and Their Jobs:

- **Spacebar:** The longest key at the bottom. It puts a blank space between your words.
- **Enter (or Return) Key:** It has a bent arrow. It moves the cursor to the beginning of the next line (like in typing) or tells the computer "OK, go ahead" (like when you type a website address).
- **Backspace Key:** It has a left arrow ←. It erases the letter to the *left* of the blinking cursor. Use it to fix typing mistakes.
- **Caps Lock Key:** When you press it once (a light may come on), every letter you type becomes a CAPITAL LETTER. Press it again to turn it off.



The Mouse: Our Pointing Tool The mouse controls a moving arrow or pointer on the screen.

Basic Actions:

- **Left Click:** Press the *left* button once. This is used to *select* an item (like an icon or a word).
- **Right Click:** Press the *right* button once. This usually opens a special menu with more options related to what you clicked on.
- **Scroll Wheel:** The wheel between the two buttons. Roll it up or down to move a page up or down without clicking.
- **Drag and Drop:** This is a two-step action. 1) *Click and HOLD* the left button on an item (like an icon). 2) *MOVE* the mouse (while holding the button) to a new location. 3) *Let go* of the button to "drop" the item there. It's like picking up a book from one table and placing it on another.

Other Input Methods

- **Touchpad:** This is a flat, sensitive rectangle on a laptop, below the keyboard. You slide your finger on it to move the cursor. You tap it lightly to "click."

- **Touchscreen:** This is a screen you can touch directly with your finger (or a special pen) to give commands. You tap icons to open them, swipe to turn pages, and pinch two fingers together to zoom out. Tablets and smartphones use touchscreens.
- **Scanner:** This is a device that takes a physical photograph or paper document and creates a perfect digital copy of it inside the computer. It's like taking a photo with a camera, but it's made specially for flat papers.

EVALUATION

- Which key do you press to make a space between two words? (The Spacebar).
- You have typed a website name and are ready to visit it. Which key do you press to tell the computer "Go"? (The Enter key).
- How do you move up and down on a long webpage using a mouse? (By rolling the Scroll Wheel).
- Your teacher has a paper drawing she wants to put on the computer. Which input device should she use? (A Scanner).
- To open an app on a tablet, do you use a mouse or your finger? (Your finger, because it has a Touchscreen).

ASSIGNMENT

- (Drawing Practical) Draw the outline of a computer mouse. Label the "Left Button" and the "Right Button."
- (Written) Look at a keyboard (at home or in the lab). Write down the first six letters you see on the top row of letter keys.
- (Oral/Written) Explain the difference between using a mouse and using a touchpad. (A mouse is a separate device you move on the desk; a touchpad is built into the laptop and you slide your finger on it).

WEEK 9: OUTPUT DEVICES.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Explain the function of a monitor.
- Differentiate between soft copy and hard copy.
- List the two main types of printers (Inkjet and Laser).
- Describe the use of a projector.

CONTENT

The Monitor: Showing Us Information The monitor is the screen that displays everything from the computer—your words, pictures, games, and videos.

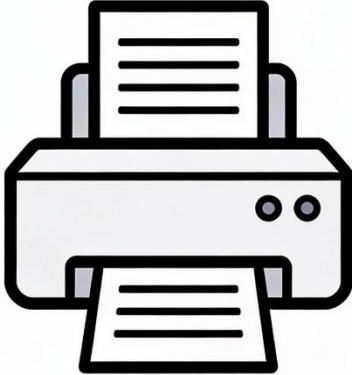
- **Soft Copy:** The information you see on the monitor is called a *soft copy*. It is "soft" because it is not a physical object; you cannot hold it in your hands. It exists only on the screen. For example, a story you are reading online is a soft copy.
- **Resolution:** This tells us how sharp and clear the picture on the monitor is. *High resolution* means the picture is very clear and detailed, like looking through a clean window. *Low resolution* means the picture might look blurry or blocky.

The Printer: Giving Us a Physical Copy The printer takes the soft copy from the computer and puts it onto paper.

- **Hard Copy:** The printed paper is called a *hard copy*. It is "hard" because it is a physical object you can touch, hold, fold, and carry. Your printed homework is a hard copy.

Main Types of Printers:

- **Inkjet Printer:** These printers use liquid ink, which comes in small cartridges. They are very good at printing colourful photographs and pictures because the colours can blend smoothly.
- **Laser Printer:** These printers use a fine powder called "toner." They are usually faster than inkjet printers and are very good for printing lots of black and white text documents, like class notes or letters. The print is also very sharp.

Inkjet Printer	Laser Printer
	

Speakers and Headphones: Giving Us Sound

- **Speakers:** These let the sound from the computer come out so that everyone in the room can hear it. Good for watching a movie with family or listening to music with friends.
- **Headphones (or Earphones):** These are worn on your head or in your ears. They let only *you* hear the sound from the computer. This is useful in a quiet place like a library, or when you don't want to disturb others.

The Projector: Making the Image Giant A projector is a device that takes the image from a computer monitor and projects (throws) it onto a large white screen or wall, making it very big. Teachers use projectors in class so all pupils can see the same lesson, picture, or video clearly from their seats.

EVALUATION

- You are reading a story on the computer screen. Is that a soft copy or a hard copy? (Soft Copy).

- Which device do we use to get a hard copy of our computer work? (A Printer).
- You are in the school library doing research. You want to listen to an educational video without disturbing others. Should you use speakers or headphones? (Headphones).
- Which type of printer uses liquid ink cartridges and is great for printing family photos? (Inkjet Printer).
- Your teacher wants to show a science video to the whole class at once. Which output device will she most likely use? (A Projector).

ASSIGNMENT

- (Drawing Practical) Draw a simple printer with a piece of paper coming out of it. Write "Hard Copy" on the paper.
- (Written) In your own words, explain what a "Hard Copy" is. (e.g., "It is my work printed on paper that I can hold.").
- (Oral/Written) Name two places, apart from your classroom, where you might see a projector being used. (e.g., Cinema, Church/Mosque, Wedding hall).

WEEK 10: COMPUTER CARE AND SAFETY.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Demonstrate proper handling of computer devices.
- List materials used for cleaning computers.
- Explain basic protection against viruses.
- Describe the correct sitting posture (Ergonomics).

CONTENT

Proper Handling of Devices: Computers are delicate and expensive. We must handle them with care.

- **Carrying:** Always use *two hands* to carry a laptop or tablet. Hold it firmly from the bottom. Never carry it by the screen.
- **Placement:** Keep computers away from the edge of a table or desk so they don't get knocked off accidentally.
- **No Food/Drinks:** Keep all snacks and drinks far away from the computer area. A single spill can destroy the keyboard or cause an electric fault.
- **Gentle Use:** Press keyboard keys gently. Do not bang or hit the mouse. Treat them like the delicate tools they are.

Cleaning the Computer: Dust and dirt can make computers hot and slow.

- **General Cleaning:** Use a *soft, dry, lint-free cloth* (like a microfibre cloth) to gently wipe the screen and the computer's body. Never spray water or cleaner directly on the screen. If needed, spray a little on the cloth first, then wipe.
- **Keyboard Cleaning:** Crumbs can get stuck under keys. You can gently turn the keyboard upside down and shake it. A small, soft brush (like a clean paintbrush) can be used to brush between the keys.
- **Liquids are Dangerous:** If any liquid spills on the computer, turn it off immediately and tell a teacher or adult.



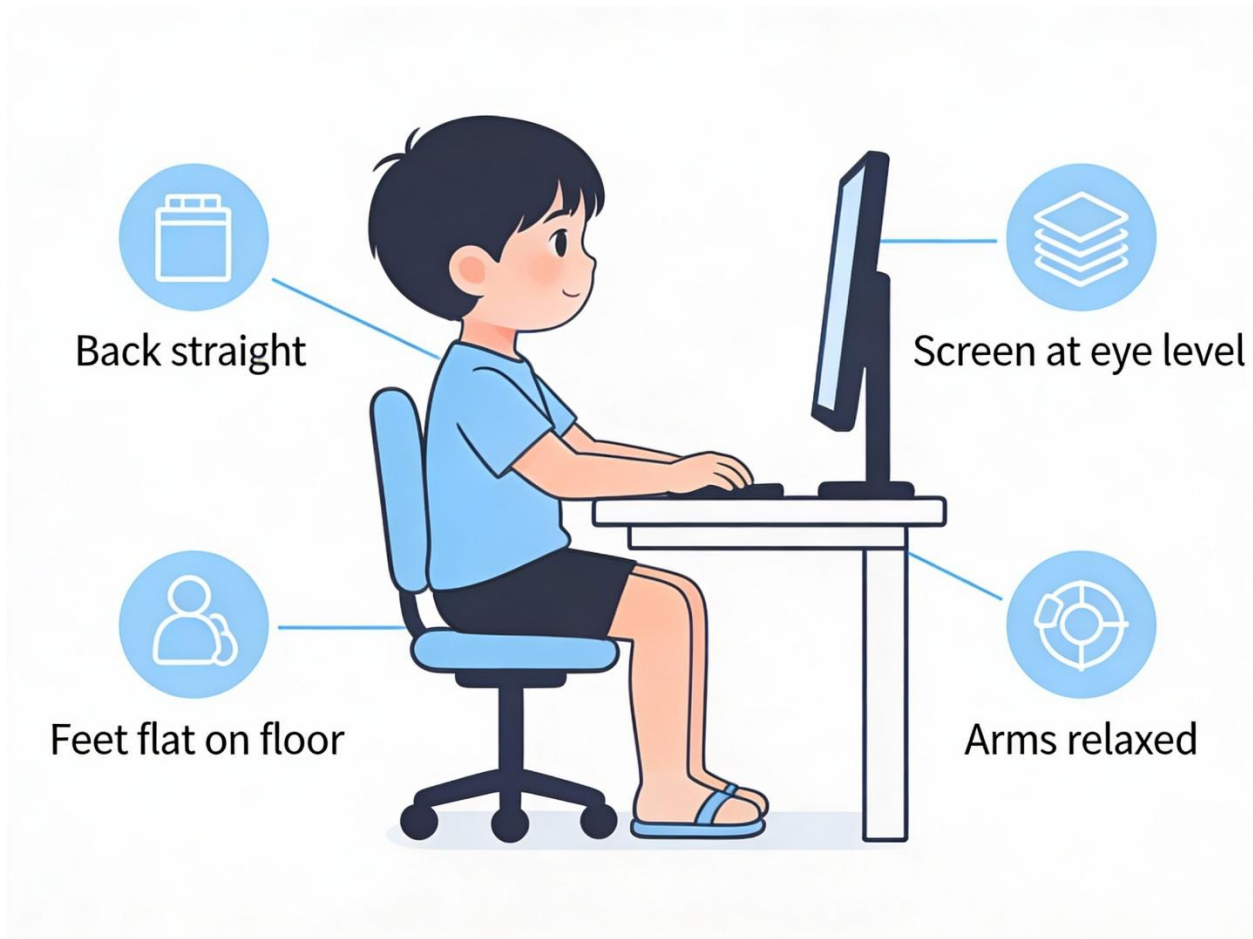
Virus Protection: Keeping the Computer Healthy A *Computer Virus* is a harmful program created by bad people. It can make your computer "sick"—slowing it down, deleting your files, or showing annoying pop-ups.

- **Antivirus Software:** This is a special program that acts like a doctor or security guard for your computer. It finds and removes viruses. It is very important to have it installed and updated.
- **Safe Habits:** Never plug in a flash drive you found on the street or from someone you don't know. It might contain a virus.

Ergonomics: Sitting Properly: Ergonomics means setting up your workspace so you can use the computer comfortably and safely, without hurting your body.

- **Posture:** Sit up straight with your back against the chair. Don't slouch or lean forward too much.
- **Feet:** Your feet should rest flat on the floor. If they don't reach, use a footrest or a stack of books.
- **Screen Height:** The top of the monitor should be at or slightly below your eye level. This stops you from bending your neck up or down.

- **Breaks:** Every 20-30 minutes, look away from the screen at something far away for 20 seconds. Get up, stretch, and walk around to rest your eyes and body.



Power Management

- **Proper Shutdown:** Always turn off the computer using the "Shut down" option from the Start menu. Never just pull the power plug from the wall while the computer is on, as it can damage the files and hardware.
- **Surge Protector:** This is a special power strip that protects the computer from sudden spikes in electricity (like during a storm) that could "burn" it. Plug the computer into a surge protector.

EVALUATION

- Why is it dangerous to have a cup of water near a laptop? (It can spill, cause a short circuit, and permanently damage the laptop).
- What should you use to clean a dusty computer screen? (A soft, dry cloth).
- What is a computer virus? (A harmful program that can damage the computer or steal information).
- Describe how you should sit when using a computer for a long time. (Sit up straight, feet flat, screen at eye level).
- Is it correct to turn off a computer by pulling its plug from the wall socket? (No, you should always use the Shut Down option from the menu).

ASSIGNMENT

- (Drawing Practical) Draw a simple stick figure sitting correctly at a computer desk. Label two good posture points.
- (Written) Write down two things you should NEVER do to a computer (e.g., Pour water on it, Hit it).
- (Oral/Written) What is the name of the special software that protects a computer from viruses? (Antivirus software). Ask a parent or guardian if your home computer has it.

WEEK 11: REVISION.

(Review of all topics covered in First Term. Practical identification of computer parts and question and answer session.)



PRIMARY 4 BASIC DIGITAL LITERACY SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	Interface Customization	Understanding computer interface. Desktop: icons, wallpaper, taskbar. Customizing wallpaper and background. Organizing desktop icons. Changing mouse pointer and cursor. Sound settings adjustment. Screen brightness control.
2	File Management I	Understanding files and folders. File types: documents, pictures, videos, music. Creating new folders. Naming and renaming files. Moving and copying files. Deleting files and Recycle Bin.
3	File Management II	Organizing files in folders. Creating folder structure. Searching for files. File properties: size, date, type. Sorting files: by name, date, size. Backup basics: saving important files. USB flash drive usage.
4	Digital Content Creation I	Introduction to creating digital content. Using Paint or drawing programs. Creating simple drawings and shapes. Using colors and brush tools. Saving artwork. Introduction to typing programs.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Digital Content Creation II	Introduction to word processing. Opening Microsoft Word or similar program. Basic formatting: bold, italic, underline, font size, colors. Typing simple documents. Saving documents with proper names.
8	Safety and Protection	Computer safety basics. Physical safety: handling devices carefully, electrical safety. Data safety: not deleting important files, saving work regularly. Password

		basics: creating strong passwords, not sharing passwords. Privacy awareness: personal information protection.
9	Safe Internet Use I	Introduction to the Internet. What is the Internet: global network. Uses: learning, communication, entertainment, information. Web browser basics: opening browser, typing web addresses. Educational websites for children. Rules for safe browsing.
10	Safe Internet Use II	Internet safety rules: never share personal information (name, address, phone, school), never meet online strangers, tell adults about uncomfortable content, use child-friendly websites only. Appropriate online behavior. Screen time limits. Parental supervision importance.
11	REVISION	Review of all topics covered in Second Term. Practical file management exercises.
12	EXAMINATION	End of Second Term Examination (Theory and Practical)
13	CLOSING	Report distribution and vacation

PRIMARY 4 BASIC DIGITAL LITERACY LESSON NOTES

SECOND TERM

WEEK 1: INTERFACE CUSTOMIZATION.

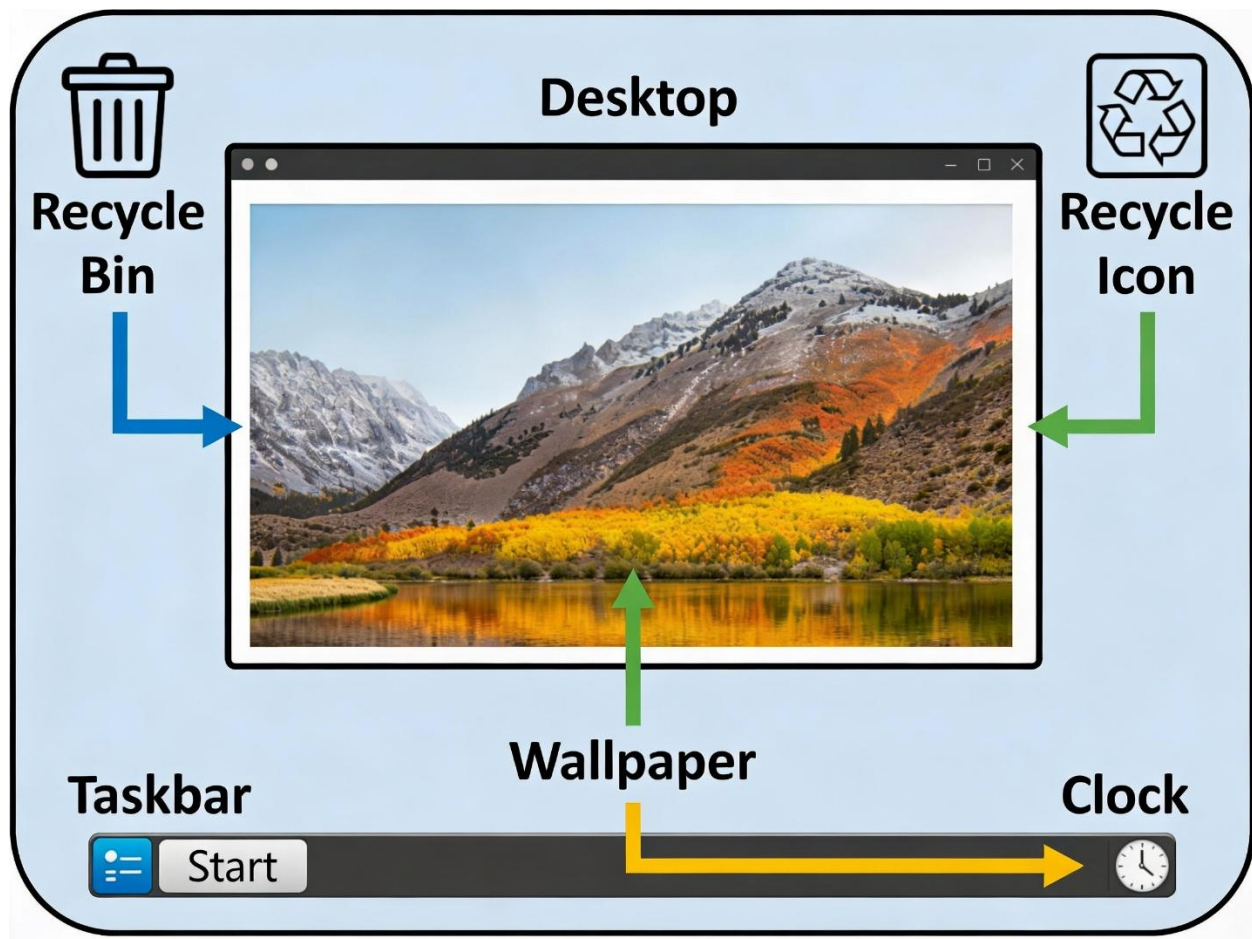
OBJECTIVES By the end of the lesson, pupils should be able to:

- Define the computer interface (Desktop).
- Identify icons, wallpaper, and the taskbar on the screen.
- Demonstrate how to change the desktop wallpaper.
- Adjust simple settings like screen brightness and sound volume.

CONTENT

Understanding the Computer Interface: When you turn on a computer and log in, the first thing you see is the *interface*. Think of it as the top of your desk at school or at home. This digital "desk" is where you keep your tools and start your work. It has a special name: the *Desktop*.

- **Desktop:** This is the main background area of your screen. It's your workspace, just like the clear space on your physical desk before you put your books on it.
- **Icons:** These are small pictures on the desktop. Each icon is like a shortcut or a label for a program, a file, or a folder. For example, the *Recycle Bin* icon looks like a wastebasket, and the *My Computer* icon might look like a computer screen. Double-clicking an icon opens it.
- **Wallpaper (or Background):** This is the picture or colour you see behind all the icons on the desktop. It's like the paint on your wall or the poster behind your desk. You can change it to a photo you like.
- **Taskbar:** This is usually a long, thin bar at the very bottom of the screen (sometimes it can be on the side or top). It is like a control panel. It has the *Start button* (the Windows logo), shows icons for programs you have open, and has a system tray with the clock, volume icon, and internet connection icon.



Customizing Your Wallpaper: You can make your computer feel personal by changing the wallpaper to a picture you enjoy—like a favourite colour, a cartoon character, or a family photo (with permission).

Step-by-Step Guide:

1. Move your mouse to an empty spot on the desktop (where there are no icons).
2. *Right-click* once with the mouse.
3. A menu will pop up. Move your mouse to click on "*Personalize*" or "*Properties*" (the word might differ slightly).
4. A new window will open with many pictures to choose from. Click on a picture you like.
5. As soon as you click it, you will see the background of your desktop change immediately!
6. You can then close the window.

Organizing Icons: A messy desktop with icons scattered everywhere can make it hard to find things, like a messy desk. We can tidy them up.

- **Moving One Icon:** Click on an icon and hold down the mouse button. While holding, drag the icon to a new, neat spot on the desktop, then let go of the button. This is *drag and drop*.
- **Auto-arranging:** To make all icons line up neatly, right-click on an empty part of the desktop. Move your mouse to "*Sort by*" and then choose an option like "*Name*" (to arrange alphabetically) or "*Date modified*" (to put the newest files first).

Adjusting Basic Settings:

- **Sound Volume:** Sometimes a video is too loud or too quiet. Look at the bottom-right corner of the taskbar for a speaker icon. *Click once* on the speaker icon. A volume slider will pop up. Click on the slider and drag it *up* to make sound louder or *down* to make it quieter.
- **Screen Brightness:** If the screen is too bright (hurts your eyes) or too dim (hard to see), you can adjust it. On a laptop, there are often special keys on the keyboard (with a sun symbol). Press the 'Fn' key and the brightness key at the same time. You can also click the battery/power icon in the taskbar; a brightness slider often appears there too.

EVALUATION

- What is the name of the main working area you see when you start the computer? (The Desktop).
- What are the small pictures on the desktop that you click to open programs called? (Icons).
- Where on the screen do you usually find the Start button and the clock? (On the Taskbar).
- To change the desktop background picture, what do you do after right-clicking on the desktop? (Click on "Personalize" or a similar option).
- Why might you need to turn down the screen brightness? (If it is too bright and hurting your eyes, or to save battery power on a laptop).

ASSIGNMENT

- (Drawing Practical) Draw a rectangle to represent a computer screen. Inside it, draw and label the "Desktop," two "Icons," and the "Taskbar" at the bottom.
- (Oral/Practical) With teacher or parent supervision, practice making the computer's volume louder and then quieter. Describe what you did.
- (Written) List two different things you can see on a computer desktop (e.g., a picture background, a recycle bin icon).



WEEK 2: FILE MANAGEMENT I.

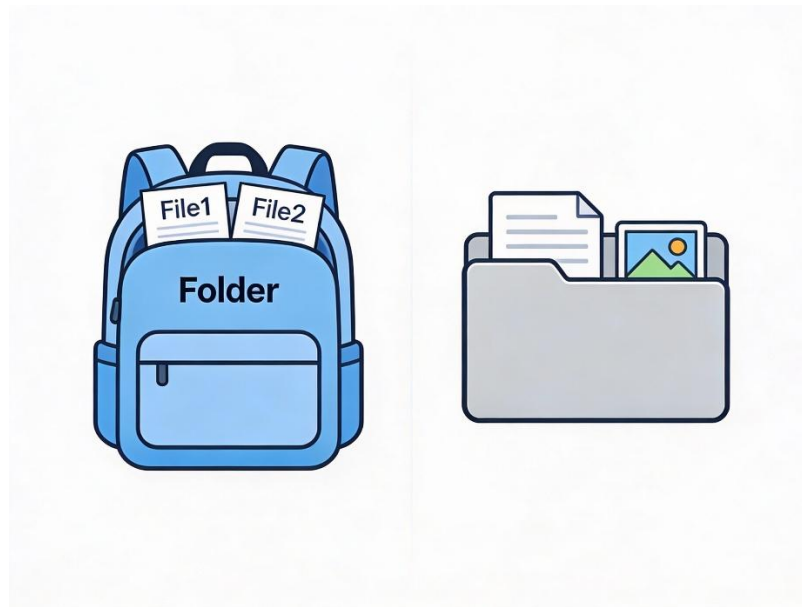
OBJECTIVES By the end of the lesson, pupils should be able to:

- Differentiate between a File and a Folder.
- Create a new folder on the desktop.
- Rename a file or folder.
- Delete a file and identify where it goes (Recycle Bin).

CONTENT

Understanding Files and Folders: Keeping things organized on a computer is like keeping your school bag tidy.

- **File:** A *file* is a single item of information. Think of it as one sheet of paper. It could be a drawing you made in Paint, a story you typed in Word, a song you downloaded, or a photograph. Every file has a name (e.g., "My Cat.jpg") and a type (like .jpg for pictures, .docx for documents).
- **Folder:** A *folder* is a container used to store and organize files. Think of it as a school bag or a box. You can put many files (sheets of paper) inside one folder to keep them together. For example, you can have a folder named "Primary 4 Homework" and inside it put files for Maths, English, and Science.



Creating a New Folder: Let's create a folder on the desktop to store your digital work.

Step-by-Step:

1. Move your mouse to an empty space on the *desktop*.
2. *Right-click* your mouse.
3. In the menu that pops up, move your cursor to "*New*." A side menu will appear.
4. From the side menu, click on "*Folder*."
5. Immediately, a new folder icon will appear on your desktop! It will have a default name like "New Folder" highlighted in blue.

Naming and Renaming Folders: A folder called "New Folder" is not helpful. We should give it a proper name.

- **Naming a New Folder:** As soon as the folder is created, its name is highlighted. You can just start *typing* the new name (e.g., "My Stories") and then press the *Enter* key.
- **Renaming an Existing Folder:** Made a mistake? Right-click on the folder icon. From the menu, select "*Rename*." The name will be highlighted. Type the new name and press *Enter*.

Deleting Files and the Recycle Bin: Sometimes we have files we don't need anymore, like old practice drawings. We delete them.

How to Delete: Right-click on the file or folder you want to remove. From the menu, click *"Delete."* The computer will ask, "Are you sure you want to move this to the Recycle Bin?" Click *"Yes."*

The Recycle Bin:

- This is a special folder on your desktop that looks like a wastebasket. When you delete a file, it isn't gone forever immediately. It goes to the Recycle Bin. This is like putting a paper in the trash can—you can still take it out if you change your mind.
- **Restoring a File:** If you delete something by accident, double-click the *Recycle Bin* icon. Find the file, right-click on it, and choose *"Restore."* It will go back to its original location.
- **Emptying the Bin:** To permanently delete everything in the Recycle Bin and free up space, right-click on its icon and choose *"Empty Recycle Bin."* After this, the files are gone for good.

EVALUATION

- What is the difference between a file and a folder? (A file is a single item like a document; a folder is a container that holds files).
- What is the usual colour of a new folder icon in Windows? (Yellow).
- When you delete a file from the desktop, where does it go first? (The Recycle Bin).
- Describe how to create a new folder. (Right-click desktop -> New -> Folder).
- If you accidentally delete your "Holiday Photo" file, how can you get it back? (Open the Recycle Bin, find the file, right-click it, and choose "Restore").

ASSIGNMENT

- (Drawing Practical) Draw the icon for a folder and colour it yellow.
- (Written) In one sentence, explain the purpose of the Recycle Bin. (e.g., "It is a temporary holding place for deleted files in case we need to recover them.").
- (Oral/Written) Name three different types of files you might find on a computer (e.g., Music file, Picture file, Video file).

WEEK 3: FILE MANAGEMENT II.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Organize files into folders.
- Search for a missing file.
- Sort files by name, size, or date.
- Explain the meaning of "Backup."

CONTENT

Organizing Files into Folders: A computer with files scattered everywhere is disorganized, like toys all over your bedroom floor. Good organization saves time.

- **Creating a System:** You can make a main folder for a big project. For example, create a folder called "*School Work 2024.*" Inside that main folder, you can create sub-folders for each subject: "*Mathematics,*" "*English Studies,*" "*Basic Science,*" and "*Cultural Arts.*"
- **Moving Files:** To put a file into a folder, use *drag and drop*. Click on the file, hold the mouse button down, drag it over the top of the folder icon, and let go. The file disappears inside the folder. Double-click the folder to open it and see the file inside.



Searching for a Missing File: What if you saved a file but can't remember where? Don't worry, the computer can help you search.

- **The Search Tool:** Look for a small *magnifying glass icon* or a search box. On many computers, it is on the taskbar next to the Start button. Click on it.
- **How to Search:** Type the name of the file you are looking for, or even just a word you remember from its name. As you type, the computer will show a list of files that match. Click on the correct one to open it.

Sorting Files Inside a Folder: When you open a folder with many files, sorting helps you find what you need quickly.

- **By Name:** Arranges files in alphabetical order (A to Z or Z to A). Good for finding a file when you know its name.
- **By Date Modified:** Puts the files you worked on most recently at the top. Useful for finding your latest homework.
- **By Size:** Orders files from smallest to largest or vice versa. Helpful if you need to find a big video file or a small document.
- **How to Sort:** Open any folder. Right-click on an empty space inside the folder window. Go to "*Sort by*" and choose your preferred option.

Backup Basics:

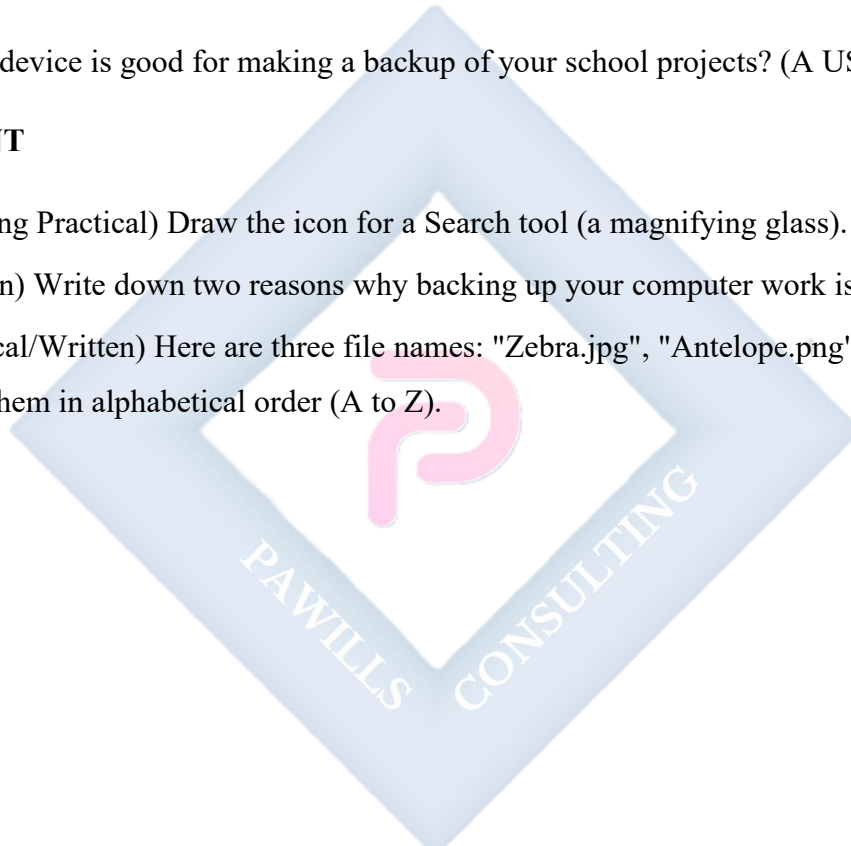
- **What is a Backup?** A *backup* is an extra, safe copy of your important files. It's like photocopying your best drawing and keeping the copy in a different, safe drawer. If something happens to the original (computer breaks, file gets deleted), you still have the backup copy.
- **Why is it Important?** Computers can sometimes fail, or you might accidentally delete something important. A backup saves you from losing your hard work forever.
- **Simple Backup Tool: USB Flash Drive:** A small, portable USB drive is a perfect tool for backing up a pupil's work. You copy your important files from the computer onto the flash drive. You can then keep the flash drive in a safe place, separate from the computer.

EVALUATION

- Why is it better to put files into subject folders instead of leaving them all on the desktop? (It keeps the desktop tidy and makes files easier to find).
- You can't find your "My Holiday" story file. What computer tool can you use to look for it? (The Search tool / magnifying glass icon).
- Name two ways you can sort files to view them in a folder. (By Name, By Date, By Size - any two).
- What does it mean to "backup" your files? (To make an extra copy and keep it in a safe place).
- Which device is good for making a backup of your school projects? (A USB Flash Drive).

ASSIGNMENT

- (Drawing Practical) Draw the icon for a Search tool (a magnifying glass).
- (Written) Write down two reasons why backing up your computer work is a good habit.
- (Practical/Written) Here are three file names: "Zebra.jpg", "Antelope.png", "Cat.gif". Write them in alphabetical order (A to Z).



WEEK 4: DIGITAL CONTENT CREATION I.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Open a drawing program (e.g., MS Paint).
- Identify tools for drawing (Pencil, Brushes, Shapes, Eraser).
- Create a simple drawing using shapes and colours.
- Save the artwork with a name.

CONTENT

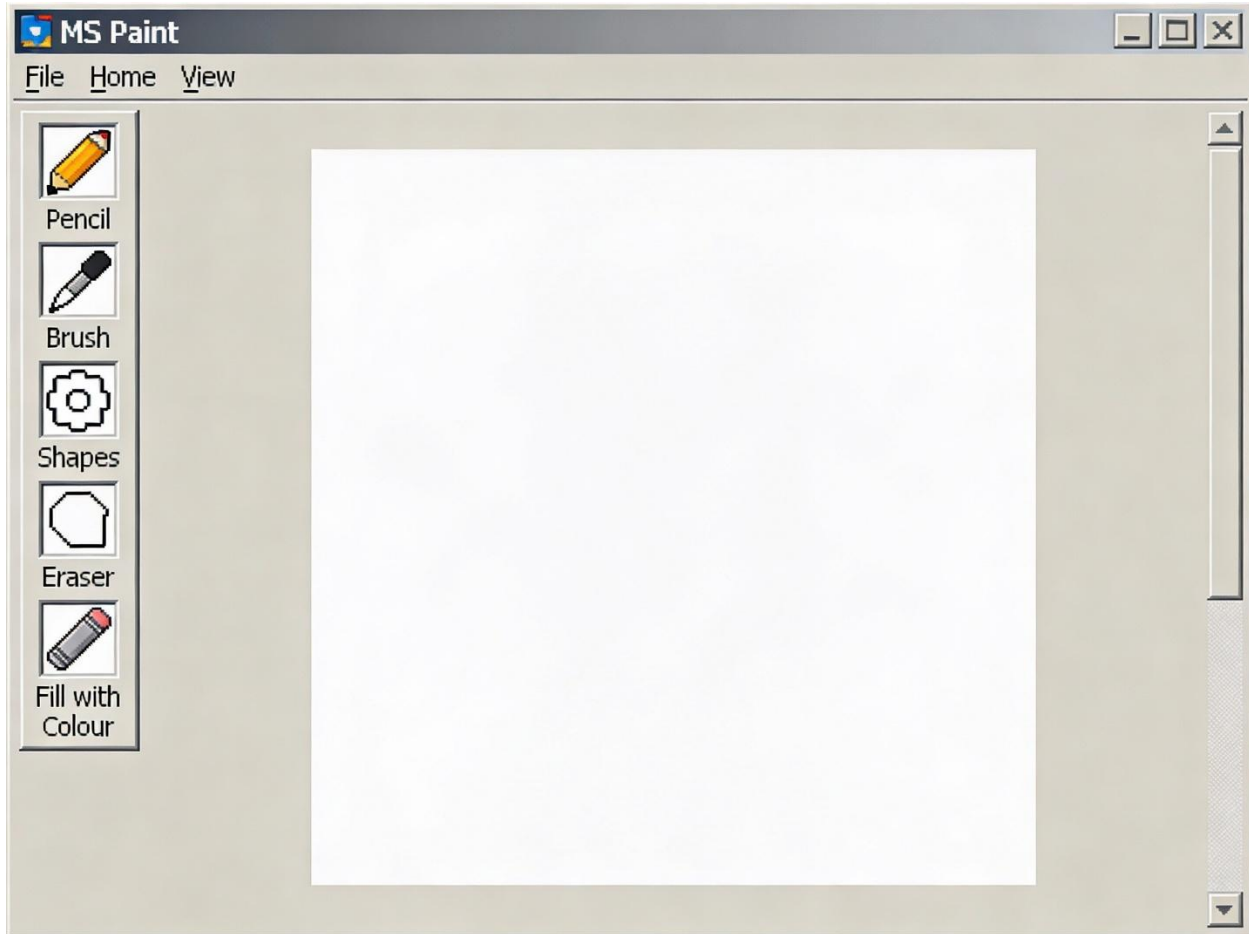
Introduction to MS Paint: MS Paint is a simple computer program that lets you draw and colour, just like using paper, pencils, and paint, but digitally! It turns your computer into a digital art studio.

How to Open Paint:

1. Click the *Start button* (Windows logo) on the taskbar.
2. Start typing the word "*Paint*".
3. The Paint app will appear in the search results. Click on it to open.

Getting to Know the Drawing Tools: When Paint opens, you will see a blank white canvas (your digital paper) and a ribbon of tools at the top.

- **Pencil Tool:** This draws thin, sharp lines, like a normal pencil. Good for outlining.
- **Brushes Tool:** This has different tips (like marker, crayon, watercolour) that draw thicker or textured lines.
- **Shapes Tool:** This lets you draw perfect shapes without freehand. You can choose rectangles, circles, triangles, stars, and arrows from a gallery.
- **Eraser Tool:** This removes parts of your drawing. It's like a rubber. You can change its size to erase small mistakes or big areas.
- **Fill with Colour Tool (Paint Bucket):** This tool looks like a spilling paint bucket. Click inside a closed shape (like a circle) to fill the entire inside with your selected colour.



Creating a Simple Drawing: Let's draw a red house with a green door and a yellow sun.

1. **Step 1: Draw the House.** Click on the *Shapes* tool. Choose the *Rectangle*. On the canvas, click and drag your mouse to draw a rectangle for the house.
2. **Step 2: Draw the Roof.** From the Shapes gallery, choose the *Triangle*. Draw a triangle on top of the rectangle for the roof.
3. **Step 3: Draw the Door.** Choose the *Rectangle* shape again and draw a smaller rectangle on the house for the door.
4. **Step 4: Draw the Sun.** Choose the *Ellipse* (circle) shape. Hold down the '*Shift*' key on your keyboard while you click and drag to make a perfect circle. Place it in the sky.
5. **Step 5: Add Colour.** Click on the *Colour Palette* to choose *Red*. Then click the *Fill with Colour* (bucket) tool and click inside the main house rectangle. It turns red! Choose *Green* and fill the door. Choose *Yellow* and fill the sun.

Saving Your Artwork: If you don't save, your beautiful drawing will be lost when you close Paint! Saving is like putting your paper drawing into your school bag to keep it safe.

Step-by-Step to Save:

1. Click the *"File"* menu in the top-left corner.
2. Click *"Save As"*.
3. A window will pop up. Choose where to save (e.g., Desktop or your *"My Pictures"* folder).
4. In the *"File name"* box, type a clear name like *"My Red House"*.
5. Click the *"Save"* button. Now your drawing is safely stored on the computer.

EVALUATION

- What is the name of a simple drawing program on many Windows computers? (MS Paint).
- Which tool would you use to correct a small mistake in your drawing? (The Eraser tool).
- You want to colour the inside of a circle blue. Which tool do you use? (The Fill with Colour / Paint Bucket tool).
- What is the most important thing to do after finishing a drawing in Paint? (Save it with a name).
- Name two shapes you can find in the Shapes tool gallery. (e.g., Rectangle, Circle, Triangle, Star).

ASSIGNMENT

- (Written) List three different tools you can find in the Paint program and what they do (e.g., Pencil: draws thin lines).
- (Drawing Practical) On a piece of paper, draw the icon for the *"Fill with Colour"* tool (a spilling paint bucket).
- (Oral/Practical) Explain to a partner the steps to open the Paint program on a computer.

WEEK 5: MIDTERM EXAMINATION.

(Assessment covering topics from Weeks 1-4)



WEEK 6: MIDTERM BREAK.

(Holiday/Rest Period)



WEEK 7: DIGITAL CONTENT CREATION II.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define Word Processing.
- Open a word processing program (e.g., Microsoft Word).
- Type a simple sentence.
- Format text using Bold, Italic, and Underline.

CONTENT

Introduction to Word Processing: *Word processing* means using a computer program to create, edit, and format text documents. It is like a super-powered typewriter. Instead of writing your letter or story by hand, you type it on the computer. This makes it neater, easier to edit, and allows you to make it look beautiful. The most common word processing program is *Microsoft Word*.

Getting Started with Microsoft Word:

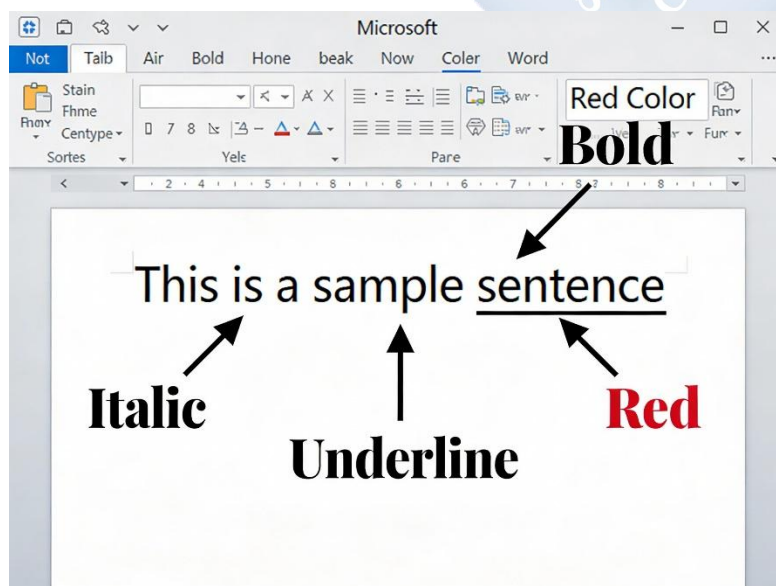
- **Opening Word:** Click the *Start button*, type "*Word*", and click on the Microsoft Word icon. A new, blank document will open.
- **Understanding the Screen:** You will see a big blank page (your digital paper) and a blinking vertical line called the *cursor*. This is where your text will appear when you start typing. At the top, you see the *Ribbon* with many tabs (like Home, Insert) full of tools.

Basic Typing Practice:

- **Typing a Sentence:** Using the keyboard, type: My name is [Your Name]. I am in Primary 4.
- **Using Important Keys:**
- **Spacebar:** Press this once after each word to make a space.
- **Enter (Return) Key:** Press this to move the cursor to the beginning of a new line. Use it when you finish a paragraph or want to start a new line.

Formatting Text: Making it Look Good: Formatting changes how your words look on the page. The main formatting buttons are on the "Home" tab in the "Font" group.

- **Bold (B):** This makes your text *thicker and darker*. It is used for *headings* or to make a word *stand out*.
- **How to:** Type a word, select it (drag your mouse over it), then click the *B* button.
- **Italic (I):** This makes your text *slanted to the right*. It is often used for *book titles* or to show *emphasis*.
- **How to:** Select text, click the *I* button.
- **Underline (U):** This puts a single line under your text. It can be used for titles or important points.
- **How to:** Select text, click the *U* button.
- **Changing Font Colour:** Next to the underline button is an "A" with a colour bar under it. Click the small arrow next to it to choose a new colour (like Red or Blue) for your selected text.
- **Changing Font Size:** Use the dropdown box with numbers (like 11, 12, 16) to make your text bigger or smaller.



Saving Your Document: Always save your work! The steps are similar to saving in Paint.

1. Click *File* -> *Save As*.
2. Choose a location (like your "School Work" folder).
3. Give it a clear name like "All About Me".
4. Click *Save*.

EVALUATION

- What do we call the activity of typing and formatting documents on a computer? (Word Processing).
- What happens to text when you use the *B* (Bold) button? (It becomes thicker and darker).
- What does the *I* (Italic) button do to text? (It makes the text slanted).
- Which key do you press on the keyboard to move to a new line? (The Enter or Return key).
- How can you change the colour of your typed words from black to blue? (Select the text, then click the Font Colour button and choose blue).

ASSIGNMENT

- (Written) On paper, write three sentences about your favourite food that you would later type into a computer.
- (Drawing Practical) Draw and label the three formatting buttons: *B* for Bold, *I* for Italic, and U for Underline.
- (Oral/Written) Give one reason why word processing is useful for a student. (e.g., "It makes my homework look neat and professional," or "I can correct mistakes easily without using a cleaner.").

WEEK 8: SAFETY AND PROTECTION.

OBJECTIVES By the end of the lesson, pupils should be able to:

- State rules for physical safety of devices.
- Explain the importance of saving work regularly (Data Safety).
- Define a Password and explain why it should be strong.
- Explain why privacy is important.

CONTENT

Physical Safety for Devices: Our digital devices are valuable and sensitive. We must protect them from physical harm.

- **Electrical Safety:** Never touch frayed, cut, or exposed wires. Always ask an adult to plug or unplug devices from wall sockets. Water and electricity are a very dangerous mix.
- **Liquid Safety:** Keep all drinks (water, juice, soft drinks) and food far away from computers, laptops, and tablets. A single spill can destroy the keyboard and internal parts, costing a lot of money to fix.
- **Handling with Care:** Always carry laptops and tablets with *two hands*, securely held. Do not run while carrying them. Close laptop lids gently. Never throw or drop any device.

Data Safety: Protecting Your Work: Data safety means making sure the information (your files, stories, pictures) on the computer is not lost or damaged.

- **Save Often:** Get into the habit of saving your work every 5-10 minutes. If there is a sudden power cut or the computer freezes, you will only lose a little bit of work instead of everything. Use the *Ctrl + S* keyboard shortcut (hold Ctrl key, press S) for a quick save.
- **Respect Others' Files:** Do not open, edit, or delete files or folders that do not belong to you unless you have permission. This includes your teacher's files, your parent's documents, or your sibling's projects.

Understanding Passwords: A *password* is a secret word, phrase, or mix of characters that you use to prove your identity. It is like a key to a locked diary or the gate to your house.

- **Why Have a Strong Password?** A weak password is easy for others to guess, like "password" or "12345". If someone guesses your password, they can get into your computer, see your private files, or even pretend to be you online.
- **How to Make a Strong Password:**
 - Make it *long* (at least 8 characters).
 - Mix *UPPERCASE* letters, *lowercase* letters, *numbers*, and *symbols* (if allowed).
 - Example: "Sunshine@74" is stronger than "sunshine".
 - Avoid using personal information like your name, birth date, or pet's name.
- **Password Privacy:** Your password is *your secret*. Never share it with friends, classmates, or anyone online. The only people you might tell are your parents or guardian.

Privacy Awareness: *Privacy* means keeping your personal life and information to yourself or only sharing it with trusted people.

- **Why it Matters Online:** On a computer used by many people (like in a lab or at home), always *log out* of your accounts when you are finished. If you leave your account open, the next person can see all your private messages and files.
- **Personal Information:** Be very careful with details like your home address, parent's phone number, or the name of your school. This information should not be shared freely, especially with people you don't know well.

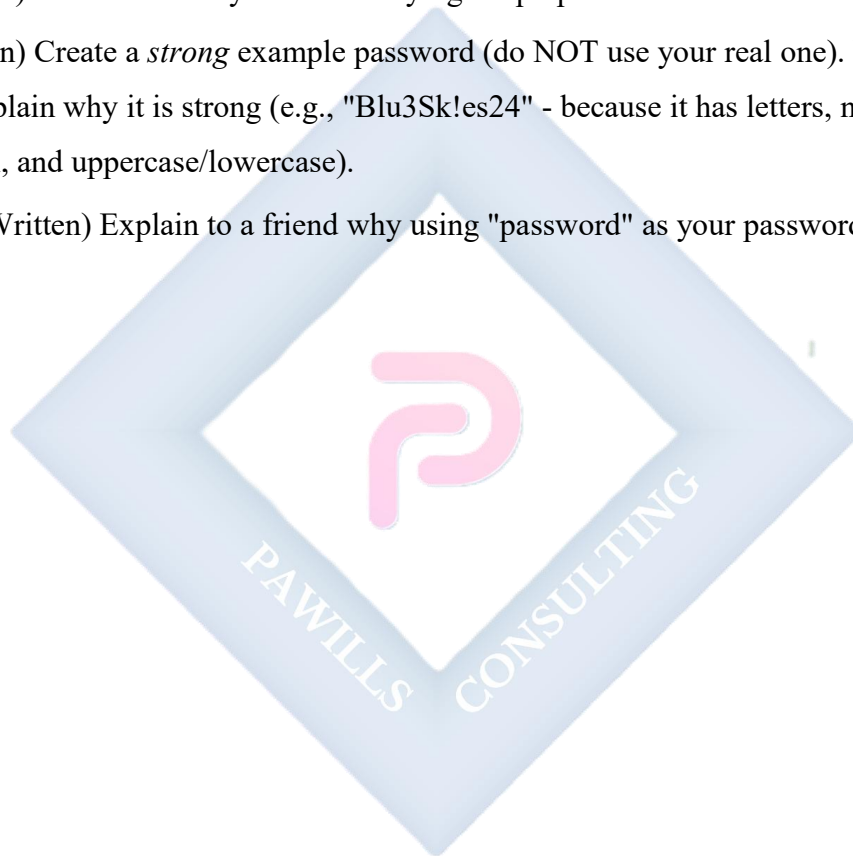
EVALUATION

- Why should you keep a cup of juice away from a laptop? (Liquid can spill and cause permanent damage to the electronic parts).
- Why is pressing *Ctrl + S* to save your work regularly a good habit? (It prevents losing all your work if the computer suddenly turns off).

- What is a password? (A secret combination of characters used to prove your identity and protect your account).
- Why is "12345" a bad choice for a password? (It is very common and easy for anyone to guess).
- Who is it okay to share your computer password with? (Your parent or guardian).

ASSIGNMENT

- (Written) Write one safety rule for carrying a laptop from one room to another.
- (Written) Create a *strong* example password (do NOT use your real one). Write it down and explain why it is strong (e.g., "Blu3Sk!es24" - because it has letters, numbers, a symbol, and uppercase/lowercase).
- (Oral/Written) Explain to a friend why using "password" as your password is a bad idea.



WEEK 9: SAFE INTERNET USE I.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define the Internet.
- List three uses of the Internet.
- Open a web browser and type a web address.
- Identify one educational website.

CONTENT

What is the Internet?: The *Internet* is a gigantic, worldwide network that connects millions of computers, tablets, and phones together. It allows these devices to share information with each other almost instantly, no matter how far apart they are. Think of it as:

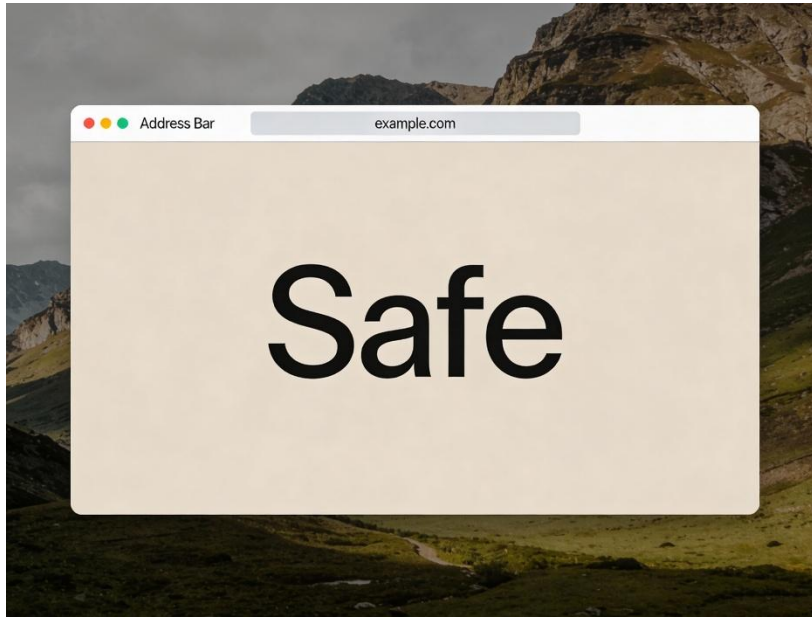
- A *global digital library* with information on every topic.
- A *huge postal system* that delivers emails, messages, and videos in seconds.
- A *massive playground* with games, videos, and music from all over the world.

Useful Things We Can Do on the Internet: The Internet is a powerful tool for learning, communicating, and relaxing.

- **For Learning (Research):** You can find information for your homework. For example, search for "facts about lions in Nigeria" or "how to solve a maths puzzle."
- **For Communication:** You can send and receive emails, make video calls to family members in another state or country, or send instant messages (with parental supervision).
- **For Entertainment:** You can watch educational cartoons, listen to music, or play safe online games during your free time.
- **For Information:** You can check the weather forecast, read news suitable for children, or find recipes for your favourite Nigerian dish.

Web Browsers: Your Window to the Internet: To visit the internet, you need a special program called a *web browser*. It is the application you open to view websites.

- **Common Examples:** *Google Chrome* (a colourful circle), *Microsoft Edge* (a blue/green wave), *Mozilla Firefox* (a red fox around a globe).
- **The Address Bar:** This is the most important part of the browser. It is a long, blank box at the very top of the window. This is where you type the *web address* or *URL* of the website you want to visit (e.g., www.google.com). When you finish typing, press the *Enter* key to go to that site.



Educational Websites for Children: The internet has many wonderful websites made just for children to learn in a fun and safe way. These sites have games, videos, and articles that match your school subjects.

- **Examples:**
- **National Geographic Kids:** For amazing facts about animals, science, and geography.
- **BBC BiteSize Primary:** For lessons and games in maths, English, and science.
- **Khan Academy Kids:** For interactive learning activities.
- **Always Ask First:** Before visiting any new website, especially for the first time, it is good practice to ask your teacher or parent if it is a safe and good site for you.

EVALUATION

- In simple terms, what is the Internet? (A worldwide network that connects computers to share information).
- Name two activities you can use the internet for, besides playing games. (e.g., Learning/Research, Sending emails, Watching educational videos).
- What is the name of the program we use to access websites? (A Web Browser).
- Where on the browser screen do you type www.bitesize.co.uk to visit it? (In the Address Bar).
- Is the internet only useful for entertainment? Yes or No? (No, it is also very useful for learning, communication, and finding information).

ASSIGNMENT

- (Drawing Practical) Draw the logo of a web browser you know (e.g., the colourful Google Chrome circle or the Firefox fox).
- (Written) List two topics you would like to search for on the internet to help with your schoolwork (e.g., "The life cycle of a butterfly," "Simple Nigerian history facts").
- (Oral) Ask a parent, guardian, or teacher to tell you the name of one website they use often for useful information. Share it with the class.

WEEK 10: SAFE INTERNET USE II.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify personal information that should not be shared online.
- Explain the concept of "Stranger Danger" online.
- State the rule for reporting uncomfortable content.
- Mention why screen time limits are important.

CONTENT

Protecting Personal Information: When you are online, you must be very careful about what you tell people, just like you wouldn't give your home address to a stranger at the market.

- **Information to Keep Private:** Never share these details in chat rooms, game comments, or with people you don't know well:
 - Your full name.
 - Your home address.
 - The name of your school or your classroom.
 - Your phone number or your parents' phone numbers.
 - Pictures of yourself, your family, or your home without your parent's permission.
- **Why?** Someone with bad intentions could use this information to find you, trick you, or trick your family.

Online "Stranger Danger": The rule "Don't talk to strangers" applies *even more strongly* on the internet.

- **Online Friends are Strangers:** Someone you only know from a game or social media is still a *stranger*. You cannot see them, verify who they are, or know their true intentions.
- **Never Agree to Meet:** If an online "friend" asks to meet you in person, you must *always* say *NO* and immediately tell a trusted adult.

- **Say No to Photos:** If someone online asks you to send them a photo of yourself, especially if you are alone or in your uniform, you must say *NO* and tell an adult.

Reporting Uncomfortable Content: Sometimes, while browsing, you might see something that makes you feel scared, confused, sad, or just "icky."

What to Do Immediately:

1. **STOP.** Do not click on anything else on that page.
 2. **CLOSE.** Close the web browser tab or window right away.
 3. **TELL.** Go straight to a trusted adult—your parent, teacher, or guardian—and tell them exactly what you saw. Do not be afraid that you will get in trouble; adults need to know to help you.
- **It's Not Your Fault:** Seeing something bad by accident is not your fault. Telling an adult is the brave and correct thing to do.

Healthy Online Habits:

- **Be Kind (No Cyberbullying):** Always be polite and respectful in your messages and comments. Never write mean, teasing, or threatening words to anyone online. This is called *cyberbullying* and it hurts people deeply.
- **Balance Your Screen Time:** Spending too many hours on screens (computer, phone, tablet) is not healthy. It can hurt your eyes, give you headaches, and take time away from important activities.
- **The 20-20-20 Rule:** Every 20 minutes, look at something 20 feet away (6 metres) for 20 seconds to rest your eyes.
- **Go Outside!** Make sure you have time every day for playing outside, reading real books, doing homework, and talking with your family *away from screens*.

EVALUATION

- Name two pieces of personal information you should never give out on the internet. (e.g., Your home address, your school name).

- If you chat with someone in an online game for a week, are they your real friend or still a stranger? (They are still a stranger).
- What are the three steps (STOP, CLOSE, TELL) you should take if you see something scary or confusing online?
- Is it okay to write a mean comment to someone in a game because they beat you? (No, that is cyberbullying and is wrong).
- Why is it important to limit the time you spend on the computer and play outside instead? (For physical health, to rest your eyes, and to have a balanced life).

ASSIGNMENT

- (Written) Make a "Private Info" list. Write down 3 things about yourself that must stay secret online.
- (Written) Complete this safety sentence: "If something I see online makes me feel uncomfortable, I will immediately tell my _____. " (parent/teacher/guardian).
- (Oral/Written) Give one good reason why playing football or jumping rope outside is better than being on the computer all afternoon.

WEEK 11: REVISION.

(Review of all topics covered in Second Term: Interface, Files, Paint, Word, and Internet Safety.
Practical exercises on creating a folder and saving a Word document.)



PRIMARY 4 BASIC DIGITAL LITERACY SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Phishing Awareness	Understanding phishing: tricking people to get information. Common phishing methods: fake emails, fake websites, suspicious links. Warning signs: spelling errors, urgent messages, asking for passwords. How to avoid phishing: don't click suspicious links, verify sender, ask adults for help. Reporting suspicious content.
2	Cyberbullying	What is cyberbullying: bullying using technology. Forms: mean messages, posting embarrassing content, excluding others online, spreading rumors. Effects: hurt feelings, fear, sadness. How to respond: don't respond to bullies, save evidence, tell trusted adults, block bullies. Being kind online. Not participating in cyberbullying.
3	Inclusivity in Digital Content Creation	Understanding diversity and inclusion online. Respecting everyone online regardless of differences. Creating content that includes everyone. Avoiding stereotypes in digital projects. Using respectful language. Celebrating diversity in digital work. Accessibility basics: content everyone can use.
4	Basics of Programming	Introduction to programming. What is programming: giving instructions to computers. Why learn programming: creates games, apps, websites, solves problems. Programming concepts: commands, sequence (order matters), algorithms (step-by-step solutions). Real-life algorithms: making sandwich, brushing teeth.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Introduction to	What is Scratch: visual programming for children, uses

	Scratch	blocks instead of typing code. Opening Scratch: online or desktop version. Scratch interface: stage, sprites, blocks palette, scripts area. Creating account (with teacher supervision). Basic navigation.
8	Coding with Scratch I	Creating first Scratch project. Adding and choosing sprites. Using motion blocks: move, turn, go to position. Using looks blocks: say, change costume, show, hide. Event blocks: when green flag clicked, when key pressed. Building simple sequences.
9	Coding with Scratch II	Using control blocks: repeat, forever, wait. Sound blocks: playing sounds, music. Making sprite interact with each other. Creating simple animations. Testing and debugging code. Saving projects.
10	Sequence in Programming	Understanding sequence: order of instructions matters. Changing order changes results. Following step-by-step instructions. Creating sequences in Scratch. Debugging: finding and fixing mistakes. Planning before coding. Sharing simple projects.
11	REVISION	Review of all topics. Complete Scratch projects. Demonstrations and presentations.
12	EXAMINATION	End of Third Term/Annual Examination (Theory and Practical)
13	CLOSING	Report distribution and vacation

PRIMARY 4 BASIC DIGITAL LITERACY LESSON NOTES

THIRD TERM

WEEK 1: PHISHING AWARENESS.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define the term "Phishing" in simple words.
- List three warning signs of a phishing message.
- Identify ways to avoid being tricked by phishing.
- Explain what to do if they receive suspicious content.

CONTENT

Understanding Phishing: *Phishing* (said like "fishing") is a trick that bad people use on the internet. Just like a fisherman uses a bait to catch a fish, these people use a fake message as "bait" to try to catch your private information. They pretend to be someone you might trust—like a popular game website, a bank, or even a friend—to fool you into giving them secrets. These secrets can be your parent's bank details, your own password, or your home address.

Warning Signs of a Phishing Message: How can you spot a fake message? Look for these red flags:

- **Bad Spelling and Grammar:** Real companies are careful. Fake messages often have mistakes, like "Congratualtions! You winn a prize!" or "Clik here."
- **A Sense of Panic or Urgency:** The message tries to scare you into acting fast without thinking. It might say: "URGENT! Your account will be blocked in 10 minutes if you don't click!" or "Act now to claim your free gift!" Real companies don't pressure you like this.
- **Asking for Secrets:** A real bank, game site, or email company will *NEVER* ask you to send your password or bank PIN in an email or message.
- **Strange or Mismatched Links:** If you hover your mouse over a link (don't click!), you might see the web address looks weird. For example, a message claiming to be from

"GTBank" might have a link like "<http://free-gifts-xyz.com>" instead of "<http://www.gtbank.com>".



How to Avoid Phishing Traps:

- **Don't Bite the Bait!** If a message feels suspicious, the safest thing is to *not click on any links* and *not download any files* attached to it.
- **Check the Sender:** Look carefully at the email address of the sender. Is it from a strange address like "security345@weirdmail.com" even though it claims to be from a famous company? That's a sign it's fake.
- **When in Doubt, Shout (for Help)!** If you are not sure if a message is real, stop. Do not reply. Show it to your parent, guardian, or teacher and ask, "Is this safe?"

Reporting Suspicious Content: If you get a phishing message, you can help stop the tricksters.

- **Do Not Reply or Engage.**
- **Tell a Trusted Adult** immediately.
- **Use the 'Report' Button:** Many email services, games, and websites have a "*Report Phishing*" or "*Report Spam*" button. Clicking this tells the website managers that someone is trying to trick people, and they can block the bad sender.

EVALUATION

- In your own words, what is phishing? (A trick where bad people send fake messages to steal private information).
- What kind of information do phishers want to steal? (Passwords, bank details, home address, etc.).
- A message says "Your game account is hacked! Click now to save it!" What is one warning sign in this message? (It creates panic/urgency; it asks for an immediate click).
- Should you ever reply to a message that asks for your password? (No, never).
- What is the first thing you should do if you receive a suspicious email offering you a free phone? (Show it to a parent/teacher and do not click anything).

ASSIGNMENT

- (Drawing Practical) Draw a warning sign (e.g., a triangle with an exclamation mark inside) and write "STOP - PHISHING" underneath it.
- (Written) Write two examples of what a phishing message might say to trick someone (e.g., "You won a free PlayStation! Send your address to get it!", "Your mummy's bank account needs verification. Send your password.").
- (Oral/Written) Explain to a friend why you should never click on a link that says "You are the 1 millionth visitor! Claim your prize!".

WEEK 2: CYBERBULLYING.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define Cyberbullying.
- List three forms of cyberbullying.
- Describe the effects of cyberbullying on children.
- Explain the steps to take when facing a cyberbully.

CONTENT

What is Cyberbullying?: *Cyberbullying* is bullying that happens using digital technology. It is when someone uses the internet, a mobile phone, tablet, or gaming console to deliberately hurt, embarrass, threaten, or scare another person, again and again. It is different from traditional bullying because it can happen anywhere, at any time—even when you are safe at home.

Forms of Cyberbullying: Cyberbullying can take many shapes:

- **Mean Messages:** Sending hurtful text messages, WhatsApp texts, or comments on social media posts (e.g., "You are ugly," "No one likes you").
- **Spreading Rumours:** Telling lies about someone online or sharing their secrets to damage their reputation.
- **Exclusion:** Intentionally leaving someone out of a group chat, online game team, or social media group to make them feel lonely and sad.
- **Impersonation:** Creating a fake profile pretending to be someone else, to post mean or embarrassing things.
- **Sharing Embarrassing Media:** Posting or forwarding an unflattering or private photo or video of someone without their permission to make others laugh at them.



Effects of Cyberbullying: Being cyberbullied hurts deeply because the mean words or pictures can be seen by many people and are hard to escape.

- **Emotional Hurt:** The victim may feel very sad, angry, afraid, lonely, or embarrassed.
- **Physical Effects:** They might have trouble sleeping, get headaches or stomach aches, or not want to eat.
- **School Impact:** They may be afraid to go to school, find it hard to concentrate in class, or their grades may drop.
- **Low Self-Esteem:** They might start to believe the mean things being said about them and feel worthless.

How to Respond: The "STOP" Strategy: If you are being cyberbullied, or if you see it happening to someone else, remember *STOP*:

- **S = Screenshot. Save the Evidence.** Take a screenshot or photo of the mean message or post. This is proof to show an adult.
- **T = Tell. Tell a Trusted Adult** immediately. This could be your parent, teacher, school counsellor, or guardian. You are not a snitch; you are being brave and asking for help.
- **O = Off. Log Off or Block.** Stop looking at the messages. Use the *block* or *report* function on the app, game, or website. This prevents the bully from contacting you again.

- **P = Pause, Don't Respond. Do NOT reply or fight back.** Bullies often want you to get upset and reply. Ignoring them takes away their power. Never become a bully yourself to get back at someone.

EVALUATION

- What makes cyberbullying different from bullying in the schoolyard? (It can happen anytime, anywhere using phones/computers, and the hurtful words/pictures can spread to many people very quickly).
- Give two examples of cyberbullying. (e.g., Sending mean text messages, posting an embarrassing photo of someone without permission).
- How might a person feel if they are being cyberbullied? (They may feel sad, scared, lonely, or embarrassed).
- Why is it important NOT to reply to a cyberbully's mean messages? (Replying often makes the situation worse and gives the bully the reaction they want).
- What does the "B" in "Block" (part of the STOP strategy) help you do? (It stops the bully from sending you more messages).

ASSIGNMENT

- (Written) Write down the four letters of the *STOP* strategy and what each one means (Screenshot, Tell, Off/Block, Pause/Don't Respond).
- (Written) Write a short, kind message you could send to a friend who you think might be feeling sad.
- (Oral) Name two digital devices where cyberbullying can happen (e.g., Smartphone, Gaming Console, Computer).

WEEK 3: INCLUSIVITY IN DIGITAL CONTENT CREATION.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define inclusivity in simple terms.
- Explain why it is important to respect everyone online.
- Identify ways to avoid stereotypes in digital projects.
- Mention one way to make digital content accessible.

CONTENT

Understanding Diversity and Inclusion: Our world is full of wonderful differences. This is called *diversity*.

- **Diversity** means that people are different from each other in many ways. Some differences you can see: skin colour, hair type, wearing glasses, using a wheelchair. Some you cannot see: religion, language, where a person is from, what they like to do.
- **Inclusion** means making sure *everyone* feels welcomed, valued, and able to participate. In the digital world, it means creating stories, games, drawings, and videos that *everyone* can enjoy and see themselves in. It's like making sure there is a seat at the table for every person.

Respecting Everyone Online: The golden rule—"Treat others as you want to be treated"—applies online too.

- **Use Kind Language:** Always use polite and respectful words in your messages, comments, and posts. Never make fun of someone because of how they look, talk, or what they believe.
- **Celebrate Differences:** When making a digital project (like a story or a poster), include characters and people from different backgrounds. Show that girls can be scientists, boys can be dancers, and anyone can be a hero.

Avoiding Stereotypes: A *stereotype* is a fixed and oversimplified idea about a particular group of people. They are often unfair and untrue.

- **Examples of Stereotypes:** "All Nigerians are good at football." "Only boys like robots." "Girls don't like video games." "Old people can't use computers."
- **Why They Are Bad:** Stereotypes put people in boxes and make them feel they have to be a certain way. They can hurt people's feelings and stop them from trying new things.
- **What to Do Instead:** In your digital creations, show a wide variety of people doing all sorts of activities. Draw a girl building a robot. Write a story about a boy who is great at cooking. Show people with disabilities as active participants.

Accessibility Basics: *Accessibility* means making things easy to use for people with different abilities, including those with disabilities. When we create digital content, we can think about how to make it accessible.

- **For People with Visual Difficulties:** Use large, clear fonts (like Arial) and good colour contrast (e.g., black text on a white background, not yellow on white). Adding a text description to images (called "alt text") helps people using screen readers.
- **For People with Hearing Difficulties:** If you make a video, add *subtitles* (the text of what is being said at the bottom of the screen). This way, someone who cannot hear the audio can still follow the story.



EVALUATION

- What does it mean to "include" someone? (To make them feel welcome and part of the group).
- Why should we use respectful language to everyone online? (Because words can hurt, and everyone deserves to be treated kindly).
- "Boys are noisy, girls are quiet." Is this a fact or a stereotype? (It is a stereotype – it is a generalisation that is not true for every boy or girl).
- How can you make a typed document easier for your grandparent with poor eyesight to read? (Use a larger font size and a clear, simple font style).
- What is one benefit of adding subtitles to a video you make? (It helps people who are deaf or hard of hearing to understand the video).

ASSIGNMENT

- (Drawing Practical) Draw a picture of a computer screen showing three different children (e.g., one in a wheelchair, one wearing a hijab, one with glasses) collaborating on a digital poster.
- (Written) Write one sentence explaining why stereotypes can be harmful. (e.g., "They make people feel they can't be themselves or try new things.").
- (Oral/Written) List two ways to be respectful when typing a message in an online class group (e.g., Say "please" and "thank you," Don't use all CAPITAL LETTERS which looks like shouting).

WEEK 4: BASICS OF PROGRAMMING.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define programming.
- List three things programming is used to create.
- Explain the term "Algorithm" using a daily example.
- State the importance of sequence (order) in programming.

CONTENT

What is Programming?: *Programming* (also called *coding*) is the process of giving a computer a set of step-by-step instructions to make it perform a specific task. A computer is very powerful, but it is not smart on its own. It needs a programmer (that could be you!) to tell it exactly what to do. The instructions are written in special languages that computers understand, like Scratch, Python, or JavaScript.

Why Learn Programming? What Can We Create?: Learning to code is like learning a superpower to build things in the digital world! Programmers create:

- **Video Games** like Minecraft, Roblox, or Football games.
- **Mobile Apps** for learning, playing, or shopping.
- **Websites** like the ones you visit for homework.
- **Animations and Cartoons.**
- **Robots** that can clean, build cars, or explore Mars!
- Programming also teaches you *logical thinking* and *problem-solving* skills, which are useful in all subjects.

Programming Concept: Algorithms: An *algorithm* is a fancy word for a simple idea. It is a clear, step-by-step plan or recipe to solve a problem or complete a task.

Real-Life Example: Algorithm for Brushing Your Teeth:

1. Pick up your toothbrush.
2. Pick up the toothpaste.

3. Unscrew the cap of the toothpaste.
 4. Squeeze a small amount of paste onto the brush.
 5. Put the brush in your mouth and brush all your teeth for two minutes.
 6. Spit out the toothpaste.
 7. Rinse your mouth with water.
 8. Rinse your toothbrush.
- If you skip a step (like forgetting the toothpaste), the algorithm doesn't work properly!
Computers need every single step.

Sequence: Why Order Matters: *Sequence* in programming means the *order* in which the instructions are given. A computer follows instructions exactly in the sequence you provide, from top to bottom.

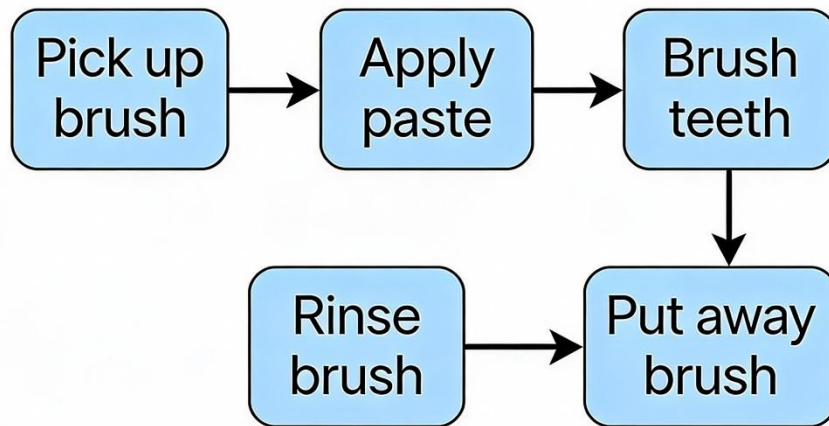
Correct Sequence Example (Making a Peanut Butter Sandwich):

1. Take two slices of bread.
2. Spread peanut butter on one slice.
3. Put the slices together.

Incorrect Sequence:

- Put the slices together.
- Spread peanut butter on one slice. (This doesn't work! The peanut butter is now on the outside!)

Conclusion: In programming, getting the sequence wrong means your program will not do what you want. It might do something silly or nothing at all!



EVALUATION

- What is programming? (Giving a computer step-by-step instructions to perform a task).
- What special name do we give to the set of instructions we give a computer? (Code or a Program).
- Name two things that are created using programming. (e.g., Video games, Websites, Mobile apps).
- What is an algorithm? (A step-by-step plan to solve a problem or complete a task).
- What happens in a program if you put the instruction "Jump" before the instruction "Stand up"? (The program will try to make you jump while sitting, which is impossible or wrong. The sequence is incorrect).

ASSIGNMENT

- (Written) Write a simple algorithm (list of steps) for an everyday activity, like *"Pouring a Glass of Water"* or *"Tying Your Shoelaces."* (Minimum of 4 steps).
- (Drawing Practical) Draw a simple robot. Write a speech bubble next to it saying: "I follow instructions in SEQUENCE!"

- (Written) Complete this sentence: "Programming is giving clear _____ to a computer." (instructions/steps).



WEEK 5: MIDTERM EXAMINATION.

(Assessment covering topics from Weeks 1-4)



WEEK 6: MIDTERM BREAK.

(Holiday/Rest Period)



WEEK 7: INTRODUCTION TO SCRATCH.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define Scratch as a programming language.
- Open the Scratch interface (online or offline).
- Identify the main parts of the Scratch screen (Stage, Sprite, Blocks Palette).
- Select a Sprite.

CONTENT

What is Scratch?: *Scratch* is a visual programming language created especially for young learners like you. Instead of typing complicated code words, you use colourful *code blocks* that snap together like LEGO bricks or puzzle pieces. By dragging and connecting these blocks, you can create your own interactive stories, games, and animations. It makes learning to code fun and creative!

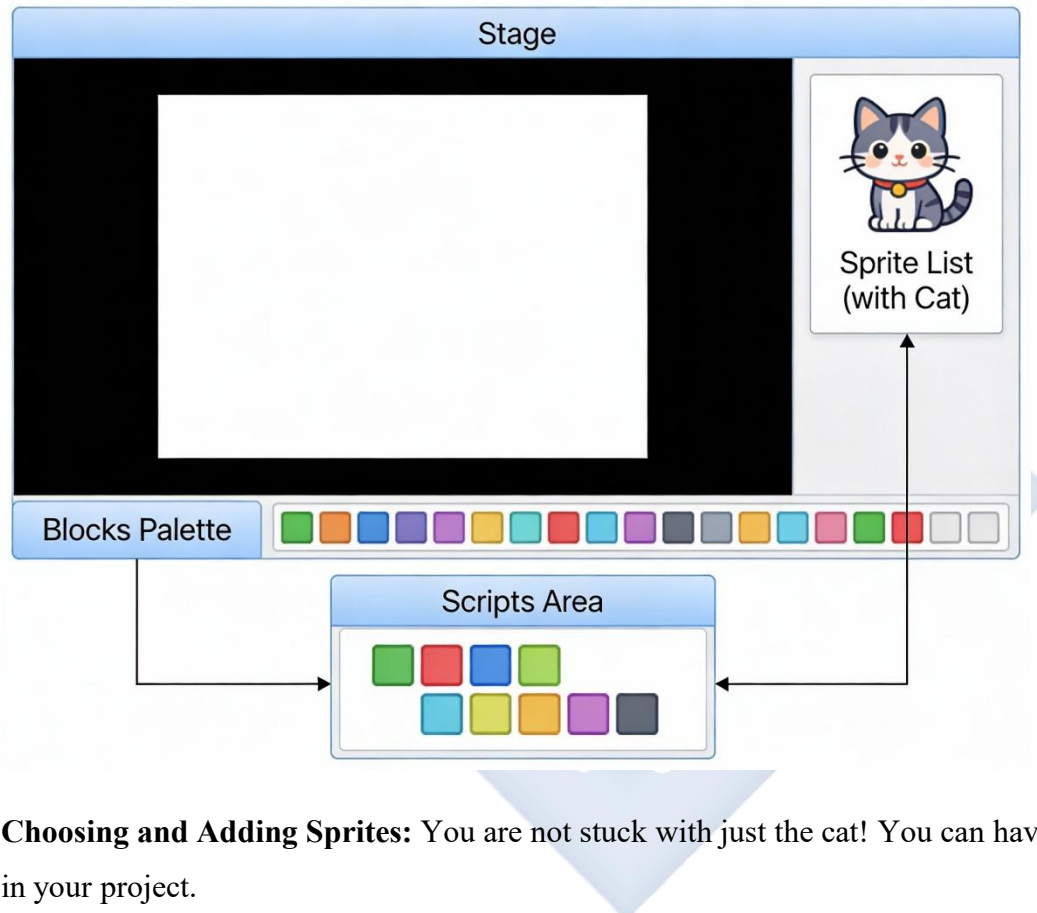
Opening Scratch: You can use Scratch in two main ways:

- **Online:** Go to the website scratch.mit.edu in your web browser. You can create an account (with teacher/parent help) to save projects online.
- **Offline:** The Scratch application can be installed on your computer. You open it by double-clicking its icon, which is an orange square with a white *cat's face* on it.

The Scratch Interface: Your Creative Workspace: When Scratch opens, you will see several important areas:

- **The Stage:** This is the big white rectangle on the left. This is where your animation or game happens—it's like a theatre stage or a TV screen. The green flag starts your program, and the red stop sign ends it.
- **The Sprite List:** Below the Stage, you see small pictures of characters. The default *sprite* (character) is the *Scratch Cat*. A sprite is any character or object that can perform actions on the stage.

- **The Blocks Palette:** This is in the middle-left, filled with colourful blocks. The blocks are grouped by colour into categories like *Motion* (blue), *Looks* (purple), *Sound* (pink), *Events* (yellow), and *Control* (orange).
- **The Scripts Area (Coding Area):** This is the big empty grey space on the right. This is where you build your program by dragging blocks from the palette and snapping them together here.



Choosing and Adding Sprites: You are not stuck with just the cat! You can have many sprites in your project.

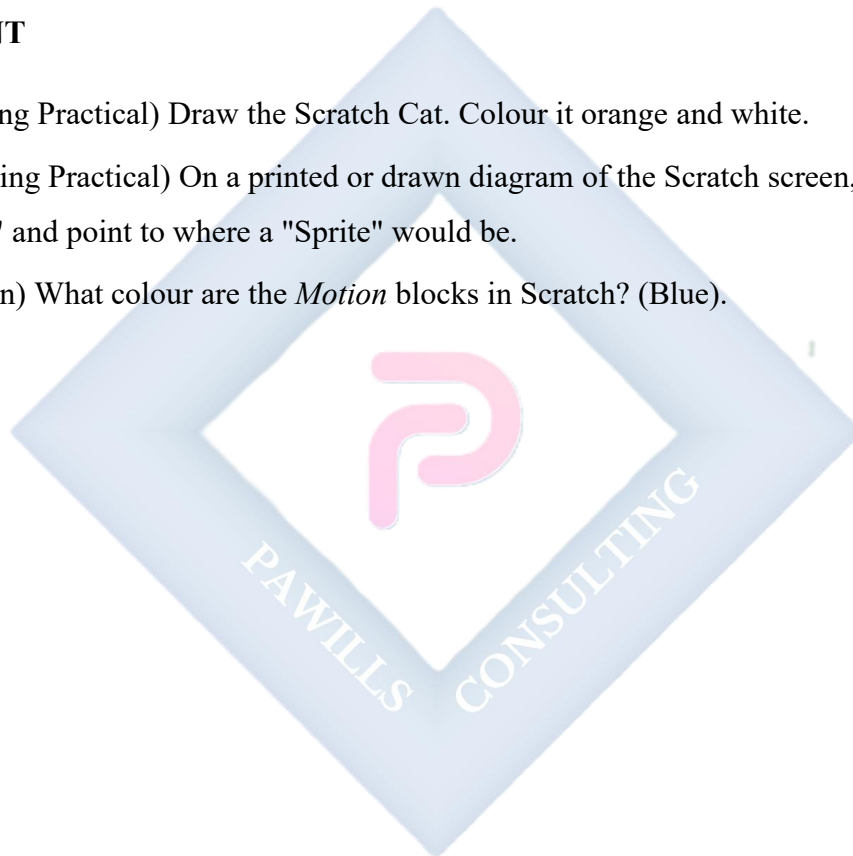
- To choose a new sprite, click the "*Choose a Sprite*" button (it looks like a cat face with a plus sign) in the bottom-right corner near the sprite list.
- A library will open with animals, people, fantasy creatures, and things.
- Click on any sprite you like (e.g., a dog, a ball, a wizard) to add it to your project.
- It will appear on the stage and in your sprite list.

EVALUATION

- Scratch is a programming language where we use _____ instead of typing words. (Colourful blocks).
- What animal is usually the default character in Scratch? (A Cat).
- What do we call the area where the action of your Scratch project happens? (The Stage).
- What is a "sprite" in Scratch? (A character or object that can be programmed to do things).
- Where do you drag blocks to build your program? (To the Scripts Area / Coding Area).

ASSIGNMENT

- (Drawing Practical) Draw the Scratch Cat. Colour it orange and white.
- (Labelling Practical) On a printed or drawn diagram of the Scratch screen, label the "Stage" and point to where a "Sprite" would be.
- (Written) What colour are the *Motion* blocks in Scratch? (Blue).



WEEK 8: CODING WITH SCRATCH I.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Add a new Sprite to a project.
- Use "Event" blocks to start a program.
- Use "Motion" blocks to move a sprite.
- Use "Looks" blocks to make a sprite speak.

CONTENT

Adding a New Sprite: Let's start a new project. Click the "*Choose a Sprite*" button (the cat with a +). From the library, pick a sprite you like—maybe a *Dog* or a *Ball*. Click on it. Now you have two sprites in your list (the Cat and your new one). Click on your new sprite in the list to select it. The blocks you drag will now control *this* sprite.

Event Blocks (Yellow): The "Start" Signals: Every program needs a way to begin. *Event blocks* are like the "Start" button for your code. They tell the program *when* to start running the blocks below them.

- The most important one is: **when green flag clicked** (yellow).
- To use it, go to the *Yellow* Events category in the Blocks Palette.
- Drag the **when green flag clicked** block into the Scripts Area.
- This will be the first block of your code stack.
- When you click the green flag above the Stage, the code attached to this block will run.

Motion Blocks (Blue): Making Things Move: *Motion blocks* control where your sprite goes on the stage.

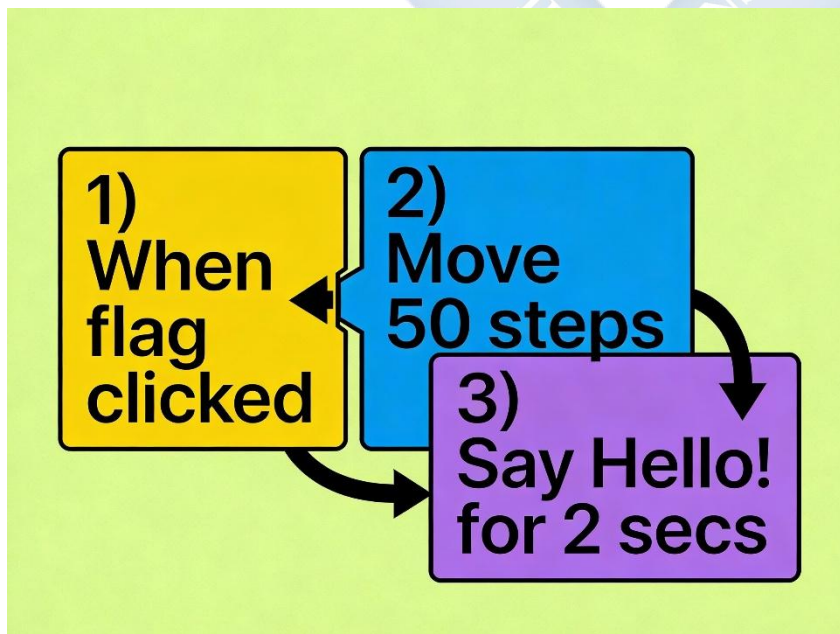
- **move 10 steps** (blue): Makes the sprite move forward in the direction it is facing. Change the number to make it move more (50) or fewer (5) steps.
- **turn right 15 degrees / turn left 15 degrees** (blue): Makes the sprite rotate or spin.
- **go to x: (0) y: (0)** (blue): Instantly sends the sprite to a specific spot on the stage. (0,0) is the very centre.

Looks Blocks (Purple): Changing Appearance and Talking: *Looks blocks* change how your sprite looks or allow it to "speak."

- **say Hello! for 2 seconds** (purple): Makes a speech bubble appear over your sprite's head. You can change the text ("Hello!") and how long it shows.
- **think Hmm... for 2 seconds** (purple): Makes a thought bubble appear.
- **switch costume to costume2** (purple): Many sprites have different "costumes" (poses or looks). This block changes the sprite's appearance, which can create simple animation (like making it look like it's walking).

Building Your First Script: Let's make the Scratch Cat walk and talk!

1. Select the *Cat* sprite from the sprite list.
2. Drag a **when green flag clicked** (yellow) block to the scripts area.
3. Go to *Motion* (blue) and drag a **move 50 steps** (blue) block. Snap it underneath the yellow block.
4. Go to *Looks* (purple) and drag a **say Hello! for 2 seconds** (purple) block. Snap it underneath the blue block.
5. Your code stack is ready! Click the *Green Flag* above the stage. The cat should move and then say "Hello!"



EVALUATION

- What colour are the blocks that make a sprite move? (Blue - Motion blocks).
- Which block do you need to start almost every Scratch program? (when green flag clicked).
- What does the say Hello! block do? (Makes a speech bubble appear over the sprite with the word "Hello").
- How do you make a ball sprite move 100 steps instead of 10? (Click on the number "10" in the move block and change it to "100").
- What happens when you click the green flag with the code you built? (The cat moves and then says "Hello").

ASSIGNMENT

- (Written/Practical) Write down the sequence of blocks to make a sprite move and speak.
- when green flag clicked
- move 30 steps
- say "I can code!" for 3 seconds
- (Drawing Practical) Draw the *Green Flag* button from the Scratch stage.
- (Written) Name two actions you can find in the *Looks* (purple) blocks menu. (e.g., "Say __", "Think __", "Switch costume").

WEEK 9: CODING WITH SCRATCH II.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Use "Control" blocks to repeat actions.
- Add sounds to a project using "Sound" blocks.
- Create a simple animation where a sprite moves and makes sound.
- Test and save a project.

CONTENT

Control Blocks (Orange): Loops and Waits: *Control blocks* let you manage the flow of your program. They can make actions repeat or pause.

- **repeat 10 (orange) Loop:** This is a "C-shaped" block. You put other blocks *inside* its mouth. It will repeat those blocks 10 times (or any number you set). Great for making a sprite walk in a square.
- **forever (orange) Loop:** This block makes the actions inside it repeat over and over again, *non-stop*, until you press the red stop sign.
- **wait 1 seconds (orange):** This block pauses your program for the amount of time you set. Useful for making a sprite wait before doing the next action.

Sound Blocks (Pink): Adding Noise!: *Sound blocks* let you add music and sound effects to your projects.

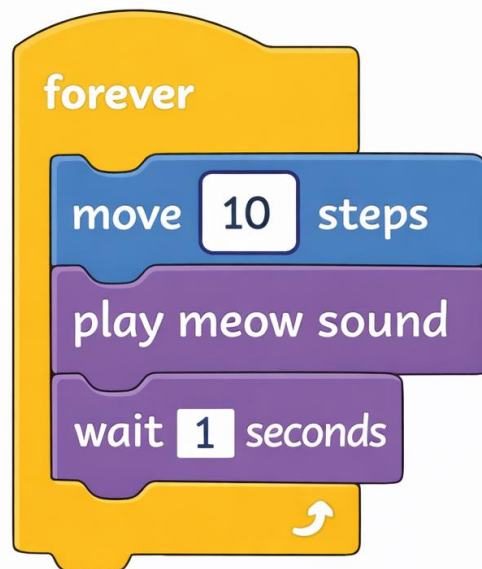
- **play sound Meow until done (pink):** Plays a sound (like "Meow") and waits for it to finish before moving to the next block.
- **start sound Meow (pink):** Starts playing a sound but immediately moves to the next block, so sounds can overlap.
- You can also click the "*Sounds*" tab near the top to record your own voice or add new sounds from a library!

Creating a Simple Animated Scene: Let's make a dancing cat that plays a drum sound!

1. **Step 1:** Start with **when green flag clicked** (yellow).

2. **Step 2:** Attach a **forever** (orange) loop below it.
3. **Step 3:** *Inside* the forever loop, add these blocks in order:
 - **move 20 steps** (blue)
 - **play drum 1 for 0.25 beats** (pink) (You can choose different drum sounds!)
 - **turn right 15 degrees** (blue)
 - **wait 0.2 seconds** (orange)
4. **Step 4:** Click the green flag! Your cat should move, make a drum sound, turn a little, pause briefly, and then repeat forever—creating a dancing animation!

Dancing Cat



Testing and Saving Your Work:

- **Testing (Running Your Code):** Always click the *Green Flag* to test if your program works as you imagined. This is called *running* your code.
- **Debugging:** If it doesn't work, check your blocks. Are they snapped together properly? Is the sequence correct? Fixing mistakes is called *debugging*.

- **Saving Your Project:** To keep your amazing animation, you must save it.
- Click *File* in the top-left corner.
- Select "*Save to your computer*".
- Give your project a descriptive name like "My Dancing Cat".
- Choose a location (like your "Scratch Projects" folder) and click *Save*.

EVALUATION

- Which Control block makes an action happen again and again without stopping? (forever).
- What colour are the Sound blocks? (Pink/Magenta).
- What is the purpose of the wait 1 seconds block? (To pause the program for one second).
- How do you stop a program that is running in a forever loop? (Click the Red Stop Sign button next to the Green Flag).
- Name one sound a sprite could make. (e.g., Meow, Pop, Drum sound).

ASSIGNMENT

- (Drawing Practical) Draw the shape of a repeat block (it looks like a "C" or a mouth).
- (Written) Write a simple script for a sprite that makes a "Meow" sound three times in a row. (Hint: Use repeat 3 and put play sound Meow inside it).
- (Oral/Practical) What menu do you click on to save your Scratch project to the computer? (The *File* menu).

WEEK 10: SEQUENCE IN PROGRAMMING.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Explain why order (sequence) is important in coding.
- Create a sequence of actions in Scratch.
- Identify and fix a simple mistake (Debug).
- Plan a simple story before coding.

CONTENT

Understanding Sequence: Order is Everything!: We learned that *sequence* means the order of steps. In programming, the computer reads and follows your code *from top to bottom, one block at a time*. If you mix up the order, you will get a different (and often wrong) result.

Example: The sequence for a robot to drink water is:

1. Pick up the cup.
2. Bring the cup to your mouth.
3. Drink.

Wrong Sequence:

- Drink. (The robot tries to drink nothing!)
- Pick up the cup.
- Bring the cup to your mouth. The sequence changed the entire outcome!

Creating a Clear Sequence in Scratch: Let's program a sprite to tell a very short story.

- **Target Story:** "The bird flew to the tree and sang a song."

Logical Sequence of Actions:

1. Start when flag is clicked. (when green flag clicked)
2. Bird says: "Time to fly!" (say Time to fly! for 2 secs)
3. Bird glides to a specific spot (the tree). (glide 1 secs to x: 100 y: 0)

4. Bird says: "I love this tree!" (say I love this tree! for 2 secs)
 5. Bird plays a chirping sound. (play sound Bird until done)
- If you put step 5 (play sound) before step 2 (say "Time to fly"), the story would be jumbled and confusing.

Debugging: Finding and Fixing Mistakes: A mistake in a computer program is called a *bug*. The process of finding and fixing bugs is called *debugging*.

- **Common Bug:** A sprite does nothing when you click the green flag.
- **Debugging Check 1:** Do you have a when green flag clicked block?
- **Debugging Check 2:** Are all your blocks snapped together properly?
- **Common Bug:** A sprite does things in the wrong order.
- **Debugging Check:** Look at the sequence of your blocks from top to bottom. Is it the logical order you planned?

The Power of Planning: Think Before You Code!: Good programmers don't just start dragging blocks. They *plan first*.

- **Unplugged Activity:** Before touching the computer, plan a simple 3-step animation on paper. Draw the sprite. Write down:
 - What should it SAY first?
 - Where should it MOVE to?
 - What SOUND should it make at the end?
- This plan is your *algorithm*. Now, in Scratch, you just need to find the blocks that match each step in your plan. This makes coding faster and easier.

EVALUATION

- What does "sequence" mean in programming? (The order in which instructions are given and followed).

- Does a computer read code from the bottom up or from the top down? (From the top down).
- What is a "bug" in computer programming? (A mistake or error in the code that causes the program not to work as expected).
- What is the process of finding and fixing bugs called? (Debugging).
- Why is it helpful to plan your animation on paper before using Scratch? (It helps you think logically about the sequence of events, making coding easier and less frustrating).

ASSIGNMENT

- (Logical Ordering) Put these Scratch blocks in the correct sequence to make sense: (a) say "Goodbye!" for 2 secs (b) when green flag clicked (c) move 50 steps (Correct Sequence: b, c, a)
- (Drawing Practical) Draw a picture of a bug (an insect) next to a computer. Write the word "DEBUG" next to it.
- (Written) In one sentence, explain why the order of blocks is so important in a Scratch program. (e.g., "If the order is wrong, the story or animation will be jumbled and not make sense.").

WEEK 11: REVISION.

(Review of all topics: Phishing, Cyberbullying, Inclusivity, Programming Basics, and Scratch Projects. Practical demonstration day where pupils show their Scratch animations.)

